

Lower Limb:

Neurovascular structures and relationships in anteromedial thigh

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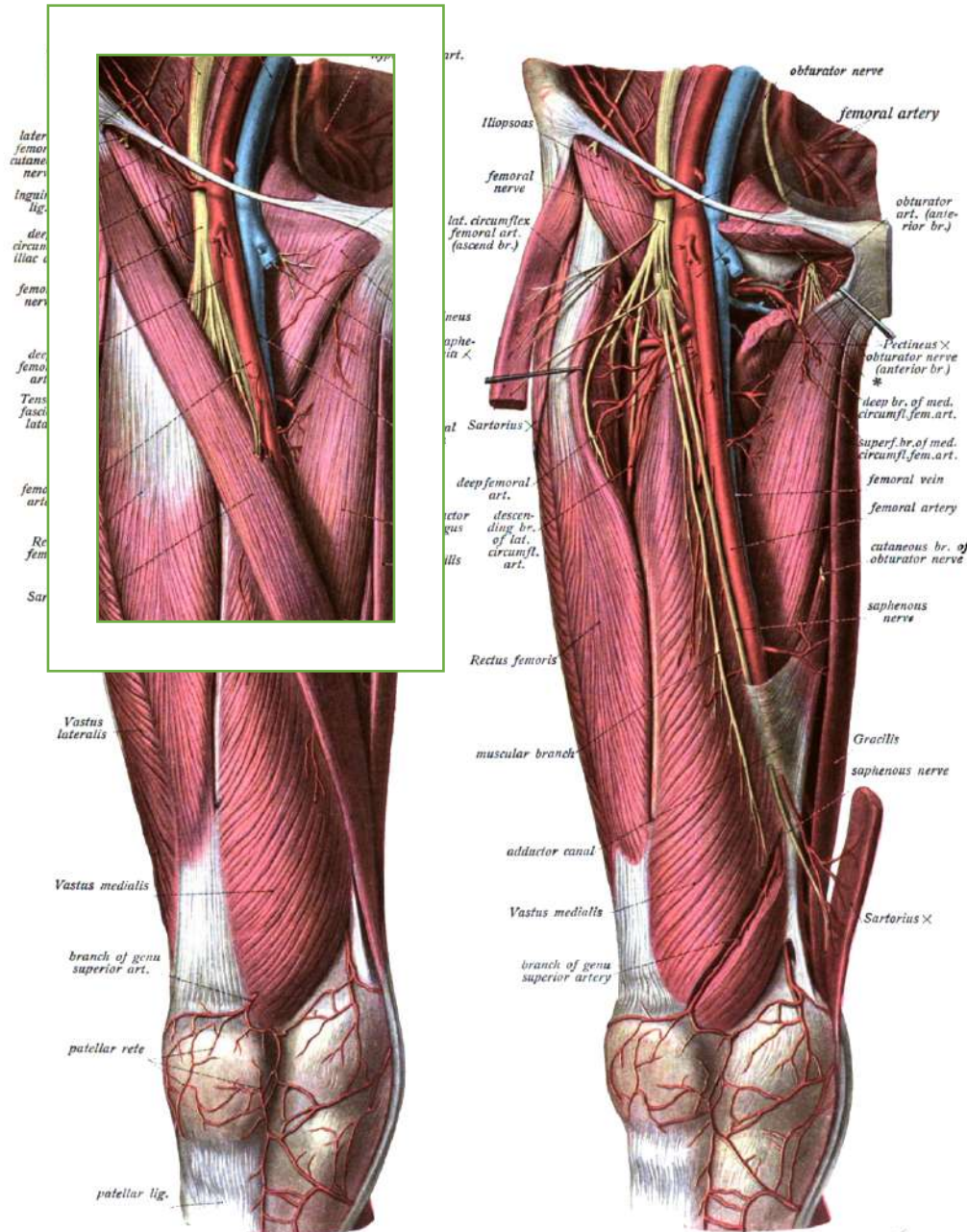


Objectives:

- ❑ Blood and nerve supply of **anterior and medial compartments** of the thigh.
- **Femoral artery & femoral nerve**
- **Profunda femoris artery**
- **Obturator nerve & obturator artery**
- ❑ Femoral triangle and femoral sheath.
- ❑ Adductor canal.



المحطات المهمة في مسار femoral vessels العظيمة



- 1) Femoral triangle
- 2) adductor canal
- 3) Adductor hiatus

المحطة الاولى Femoral Triangle



- Is a **triangular depressed** area situated in the upper part of the medial aspect of the thigh, just below inguinal ligament.

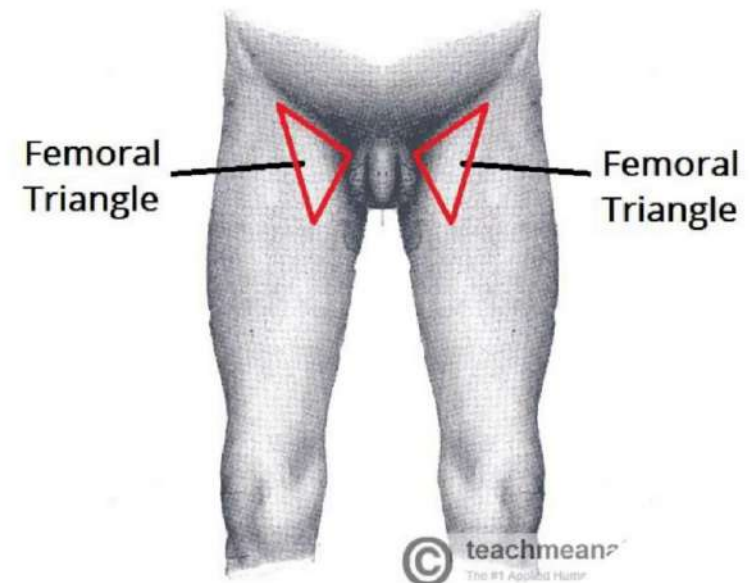
☐ Importance?

☐ Boundaries?

☐ Floor?

☐ Roof ?

☐ Content ?

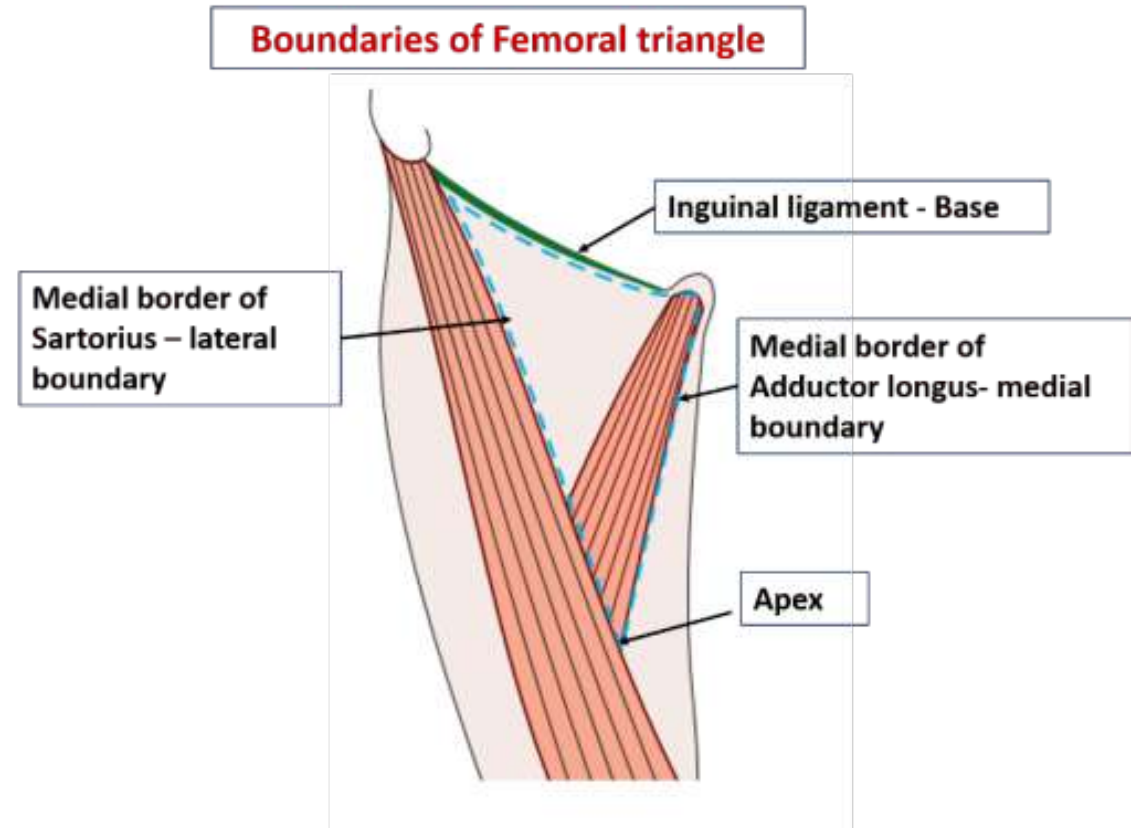




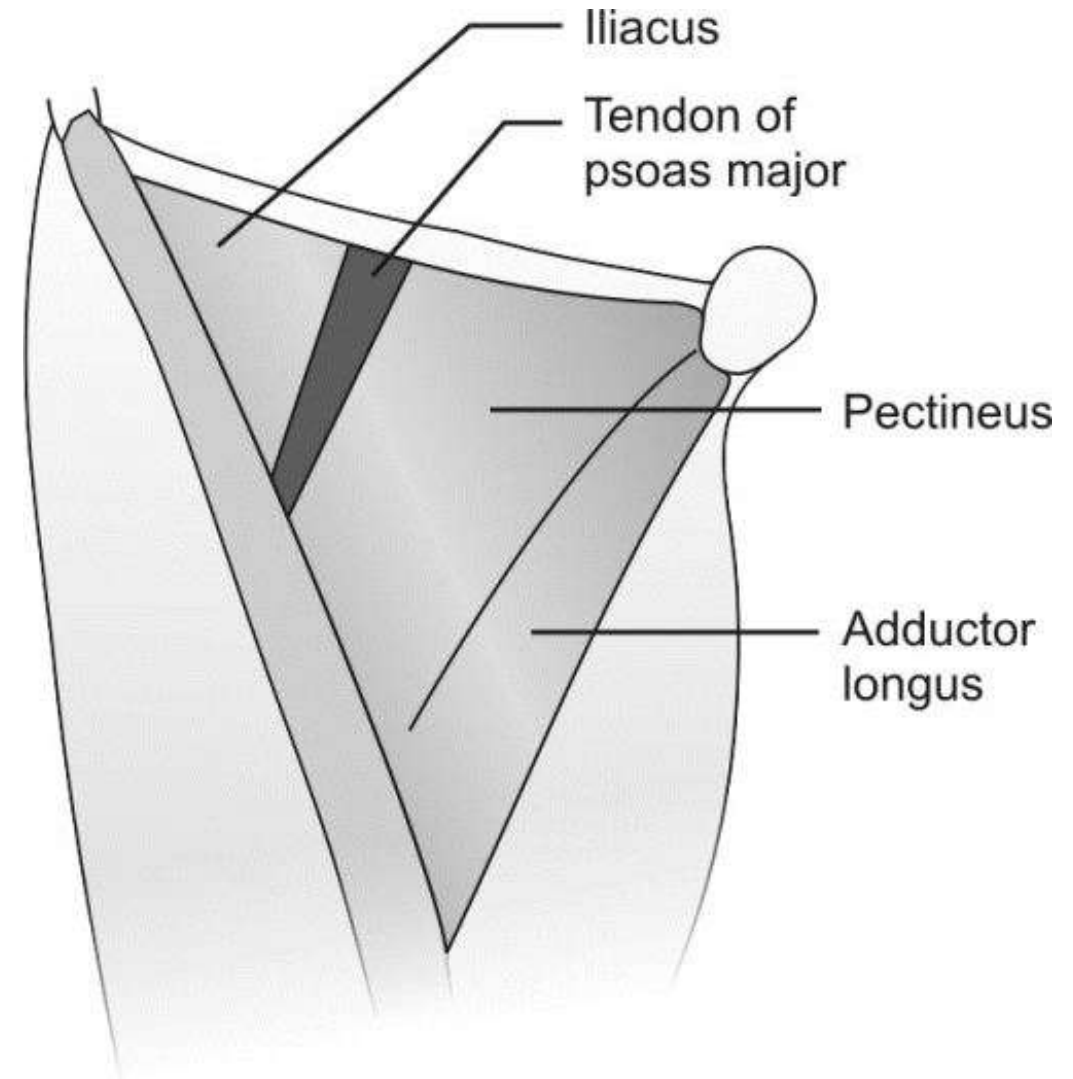
Boundaries:

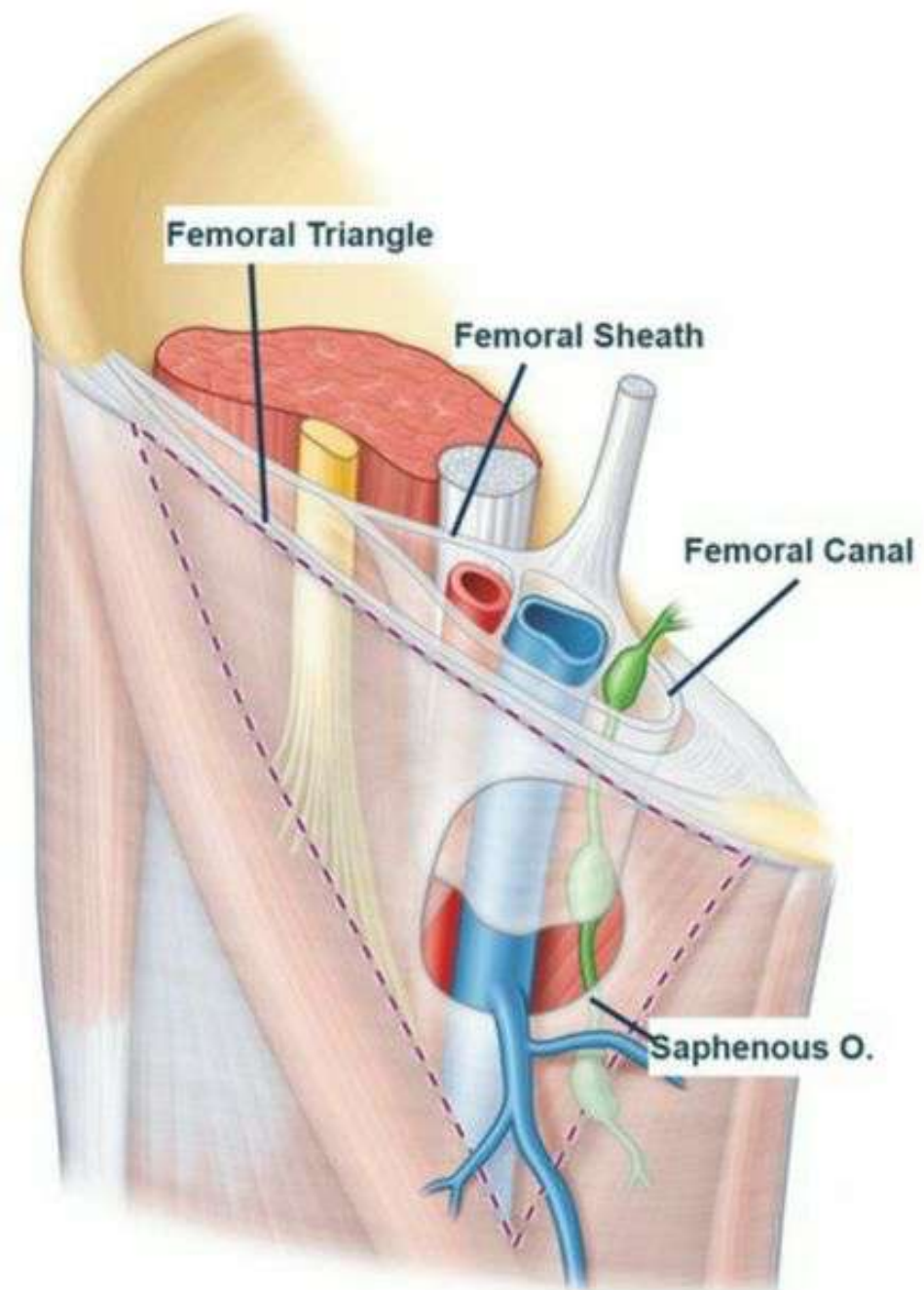
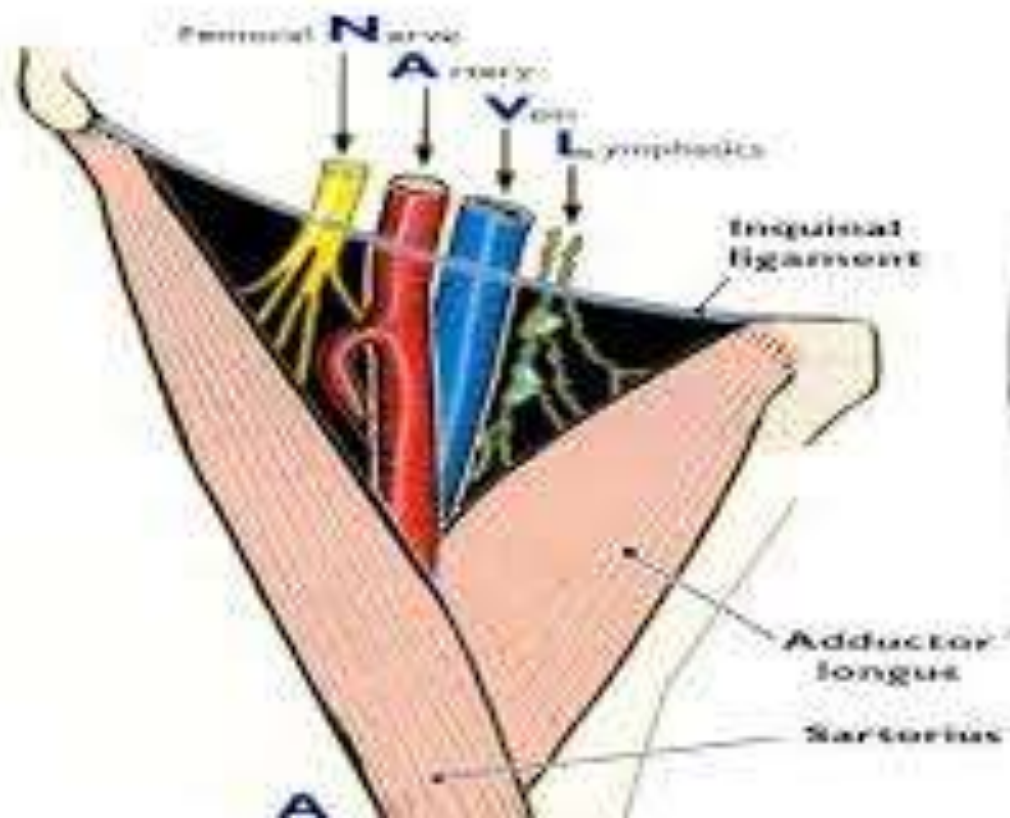
على شكل مثلث مقلوب

- ❑ Superiorly (base): by inguinal ligament.
- ❑ Laterally: by medial border of sartorius.
- ❑ Medially: by medial border of adductor longus.
- ❑ Apex: where the medial border of sartorius crosses the medial border of adductor longus.



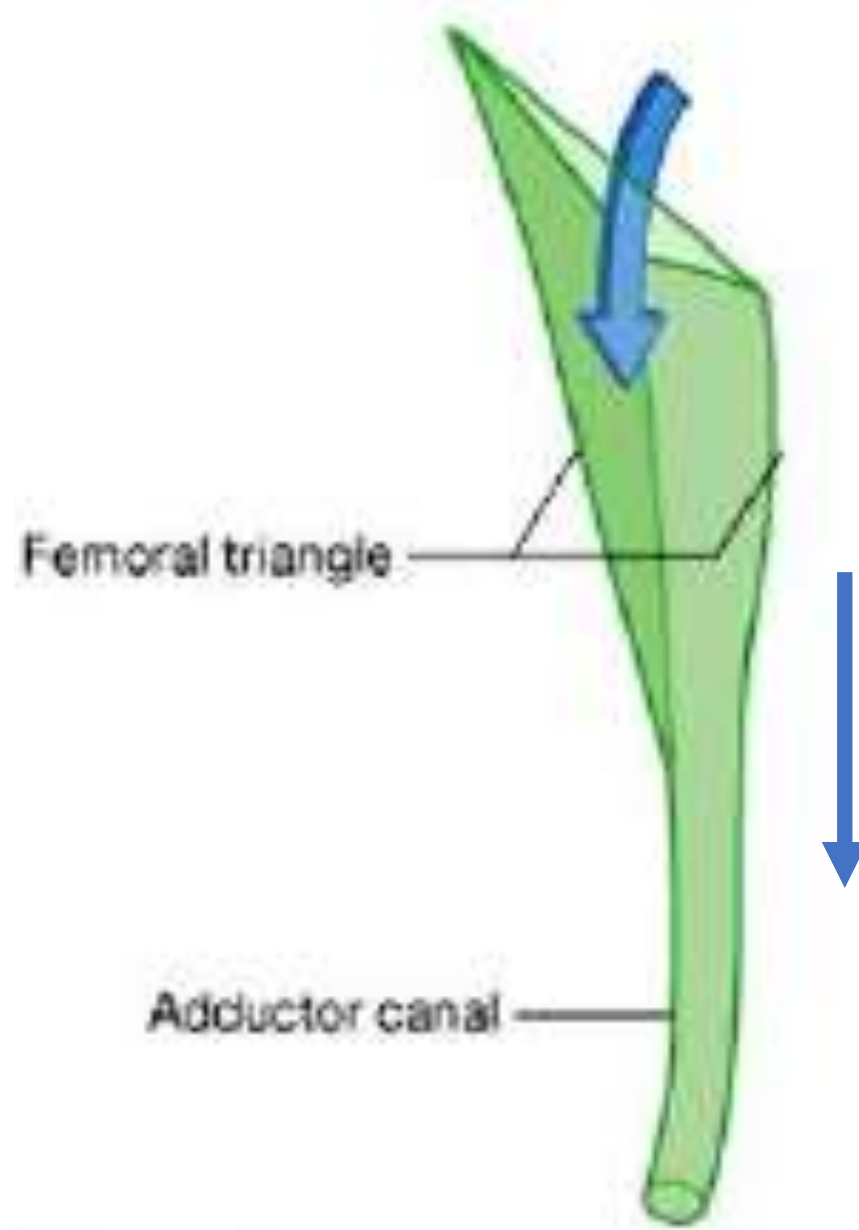
- ❑ **Floor:** gutter shaped, formed from lateral to medial by; 1) iliopsoas 2) pectineus 3) adductor longus.
- ❑ **Roof:** skin & fasciae of the thigh with *saphenous opening* in fascia lata.
- ❑ **Content** from lateral to medial:
- ❑ **Terminal part** of **femoral nerve & its branches.**
- ❑ **Femoral sheath** and its content(**femoral artery & its branches.** **Femoral vein & its tributaries.** **Femoral canal & deep inguinal lymph nodes.**)





❖ The **apex** of the femoral triangle points inferiorly & is continues with a fascial canal (adductor canal المحطة الثانية).





Important summary

1-Femoral nerve is one of the content of femoral triangle but not one of the content of femoral sheath.

2-Roof of femoral triangle is made by deep fascia of the thigh(Fascia lata): in which saphenous opening is present; so it is present in the roof of femoral triangle, just anterior to femoral vessels.

3-Femoral nerve lies lateral to femoral artery in the femoral triangle, outside the femoral sheath.



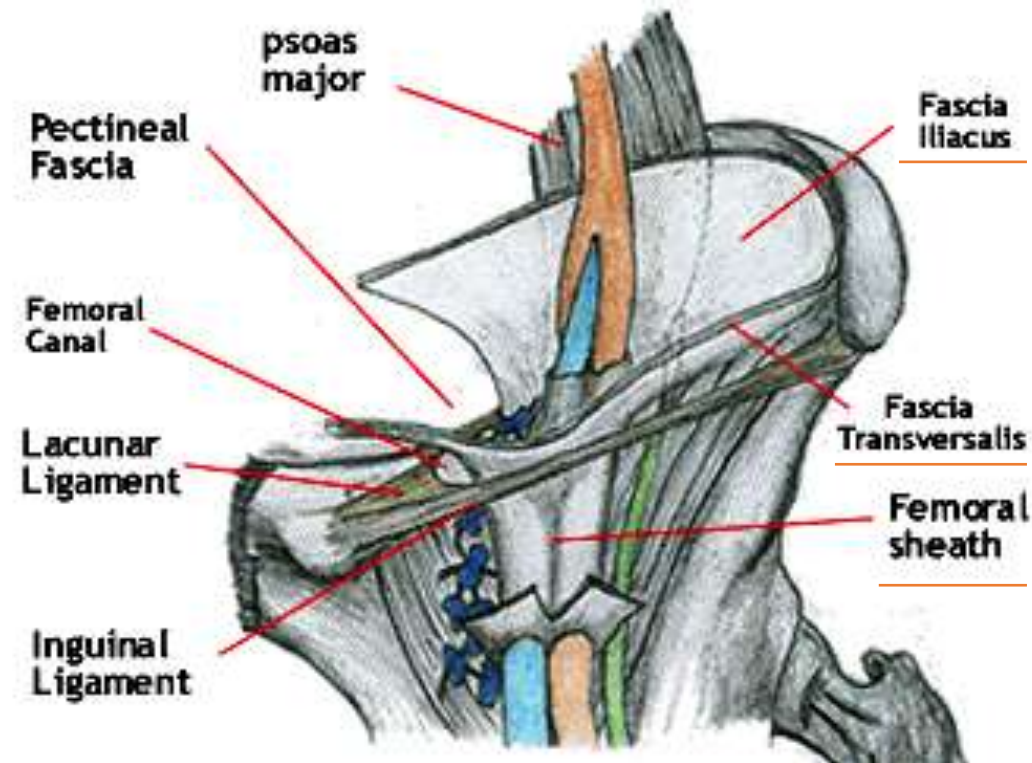
(Femoral Sheath)

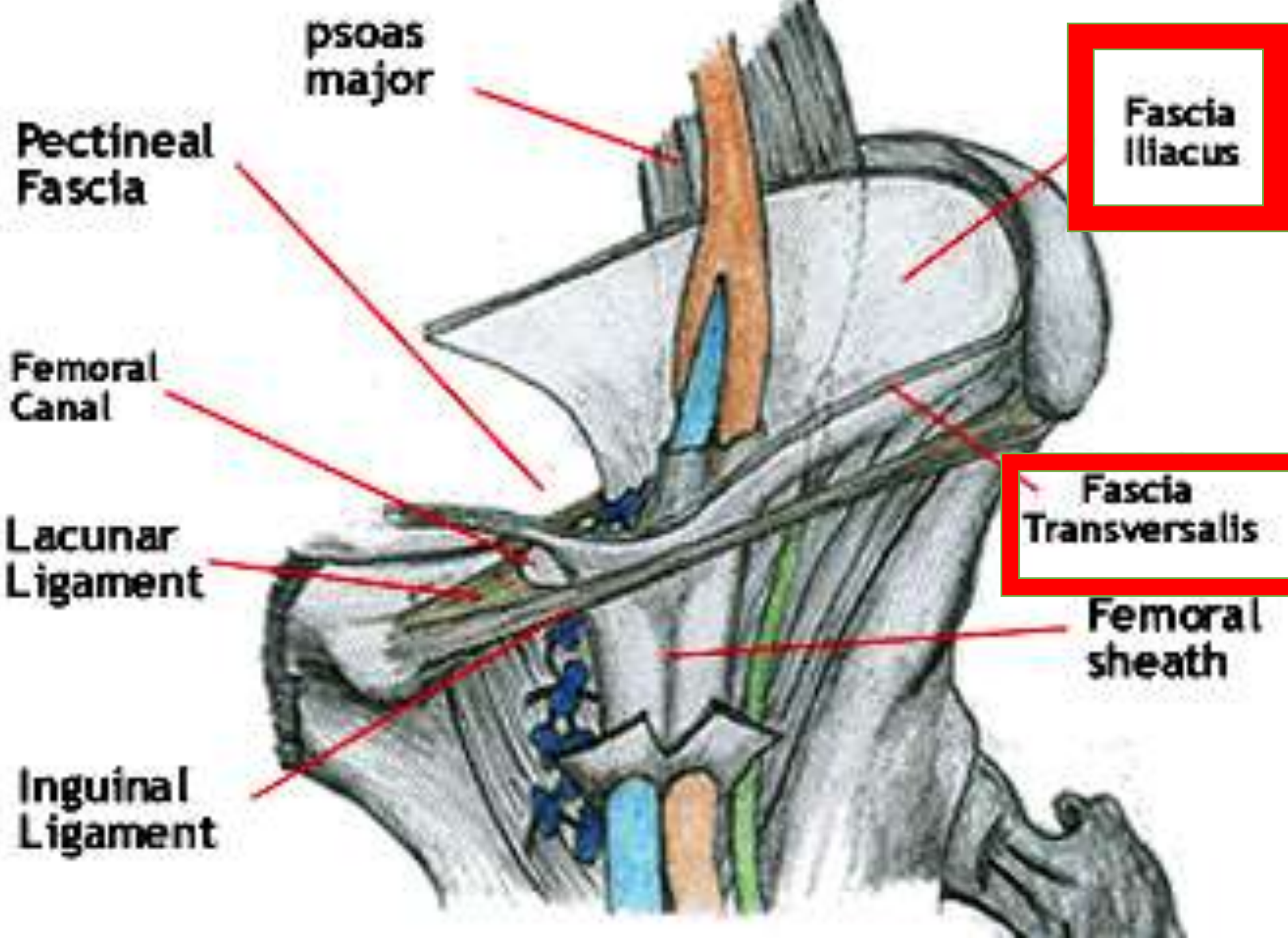


❑ **Definition:** Funnel shaped fascial sheath which extends down from abdominal walls to surround the upper 2.5 cm of **femoral vessels**.

❑ **Walls: Anterior wall:** is downward continuation of **fascia transversalis** of ant. abdominal wall.

Posterior wall: is downward continuation of **fascia iliaca** (covering **iliopsoas**) of the posterior abdominal wall.





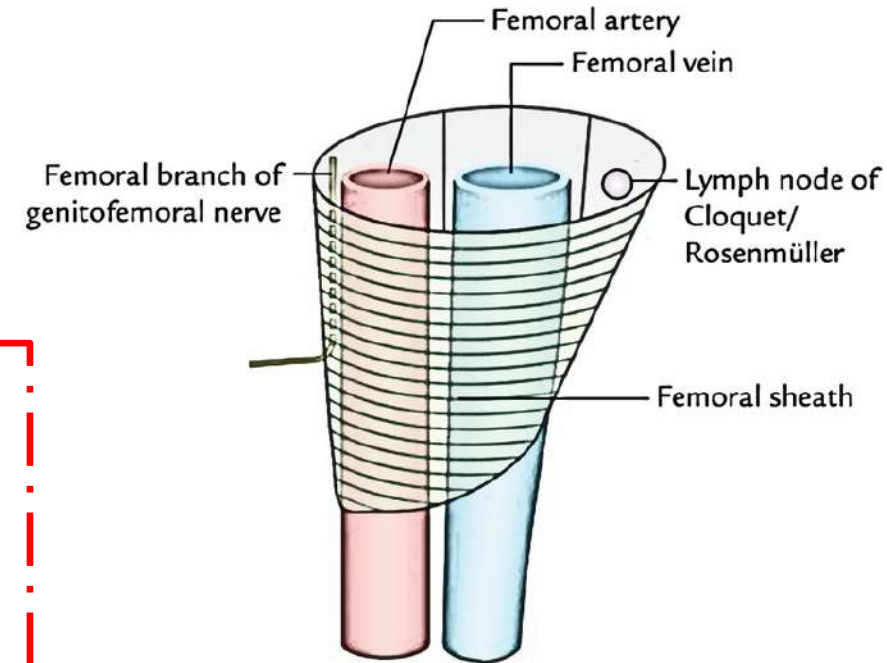
❑ **Importance:** it surrounds proximal part of femoral vessels & lymphatics for about 1 inch below inguinal ligament.

❑ **Compartments:** is divided by **vertical septa** into 3 compartments:

1. **Lateral compartment:** occupied by **femoral artery**

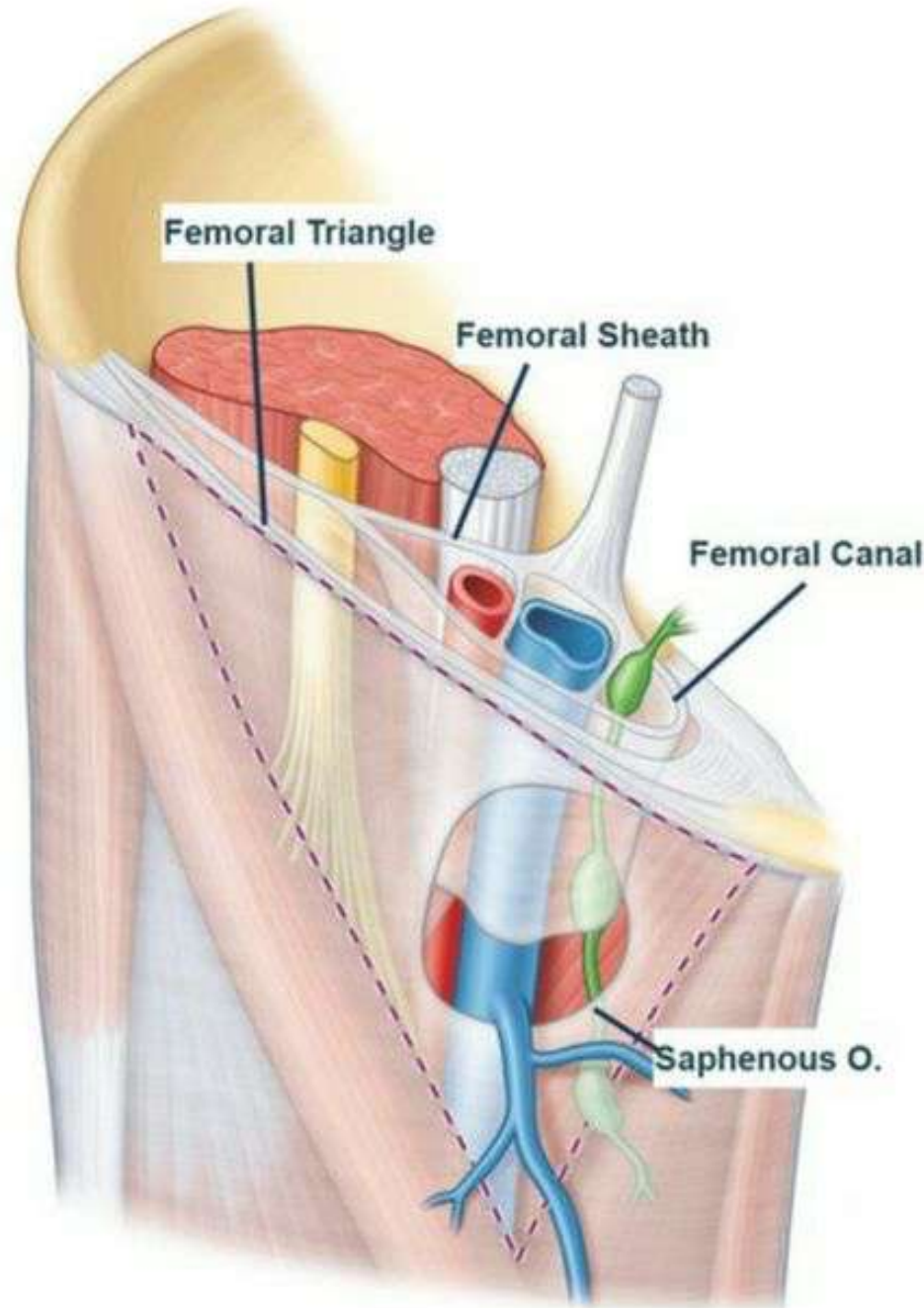
2. **Intermediate compartment:** occupied by **femoral vein**

3. **Medial compartment :** occupied by **femoral canal**.



Lateral

Medial

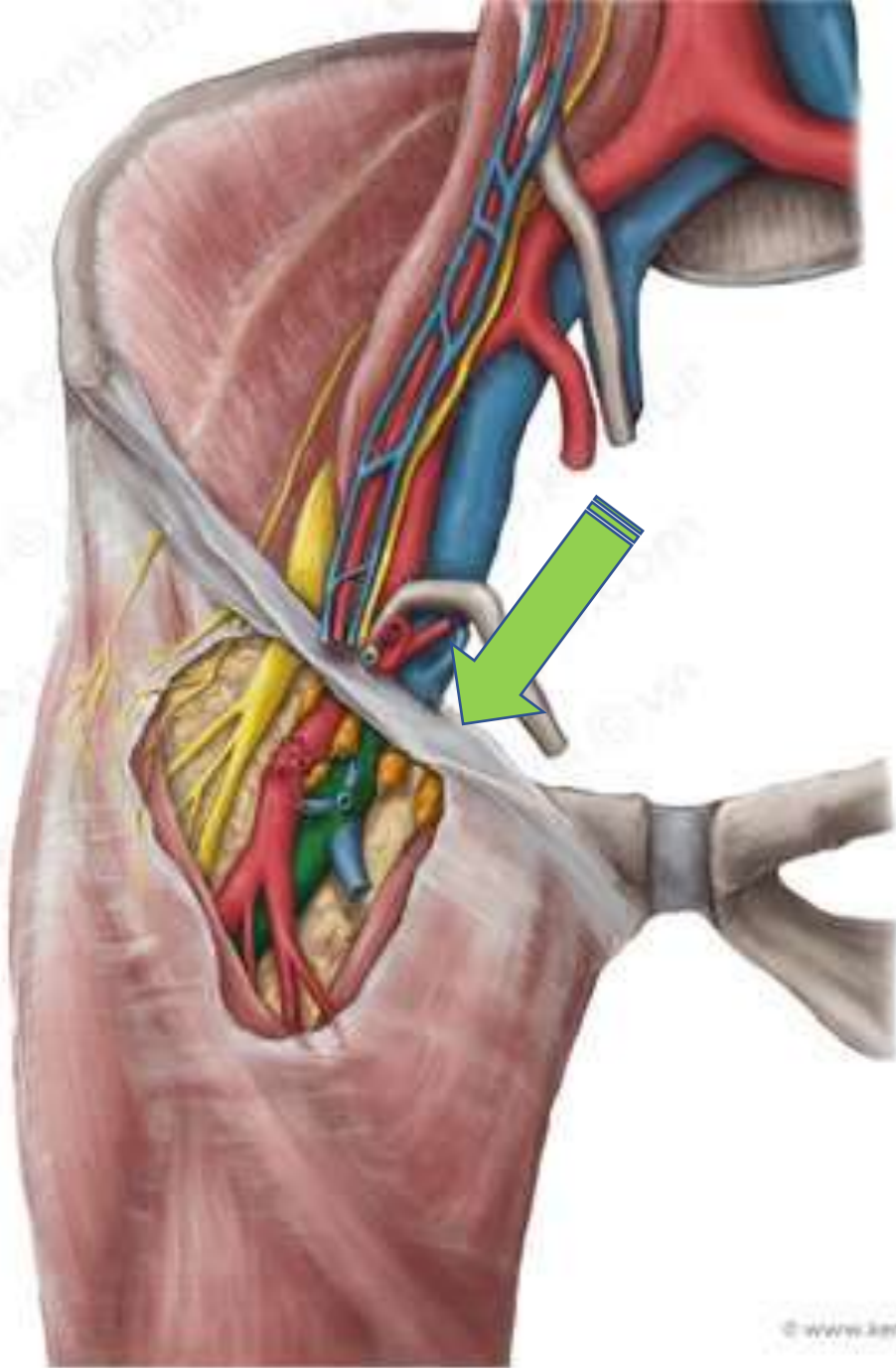


Femoral sheath

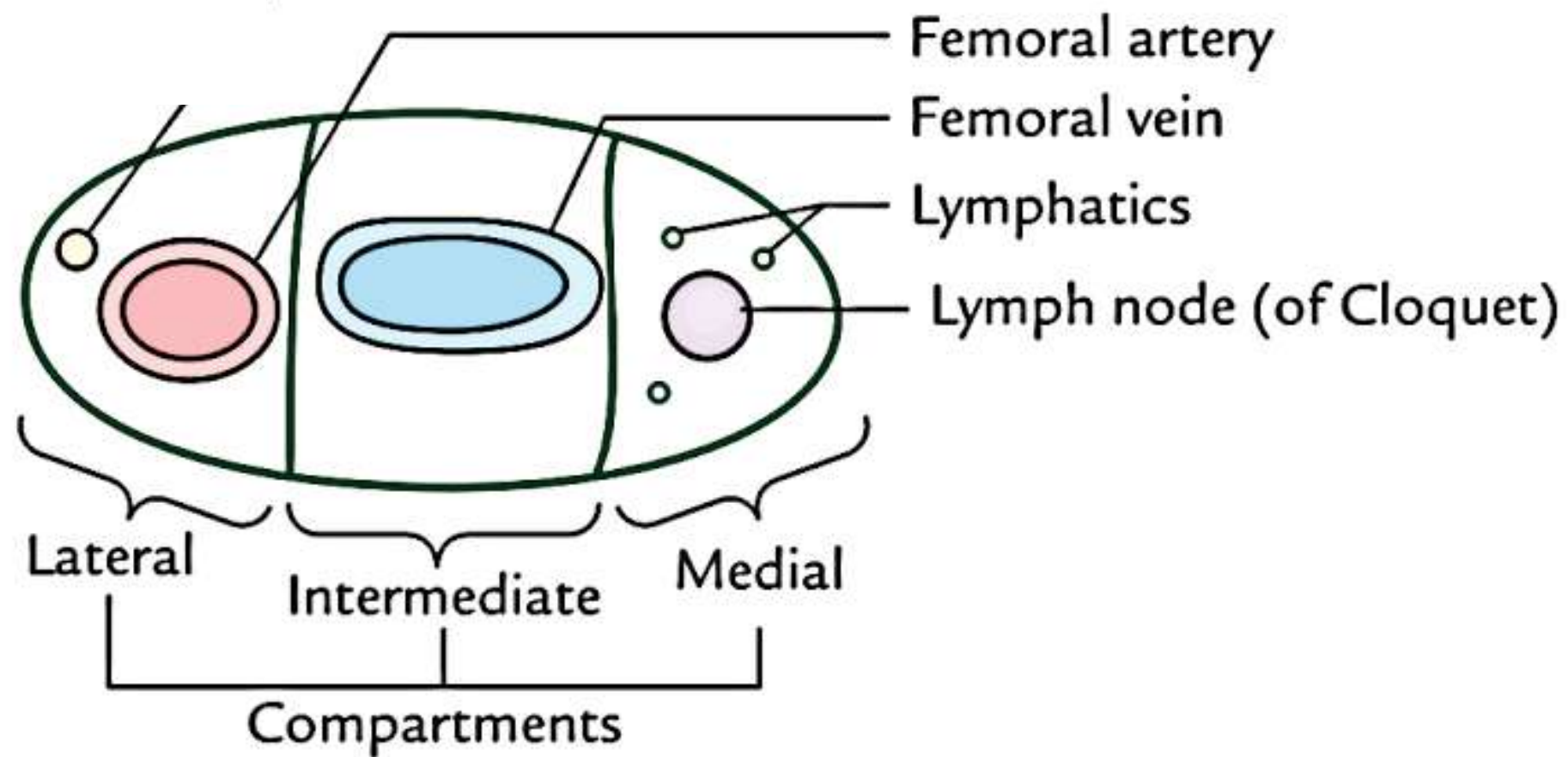
Important to know that Femoral nerve is lateral to femoral sheath and **Not** contained within the femoral sheath.

Femoral Canal مهم

- **Femoral canal:** name given for the **small medial compartment of femoral sheath** for the lymph vessels. It is about ½ inch long. is closed inferiorly.
- **Femoral ring:** is the **upper abdominal opening** of femoral canal. Which is closed by extraperitoneal fatty tissue(femoral septum).
- **Content of femoral canal: (relatively empty)**
 - Fatty connective tissue.
 - One of deep **inguinal lymph nodes**.
 - **Efferent lymph vessels** from deep inguinal lymph nodes.

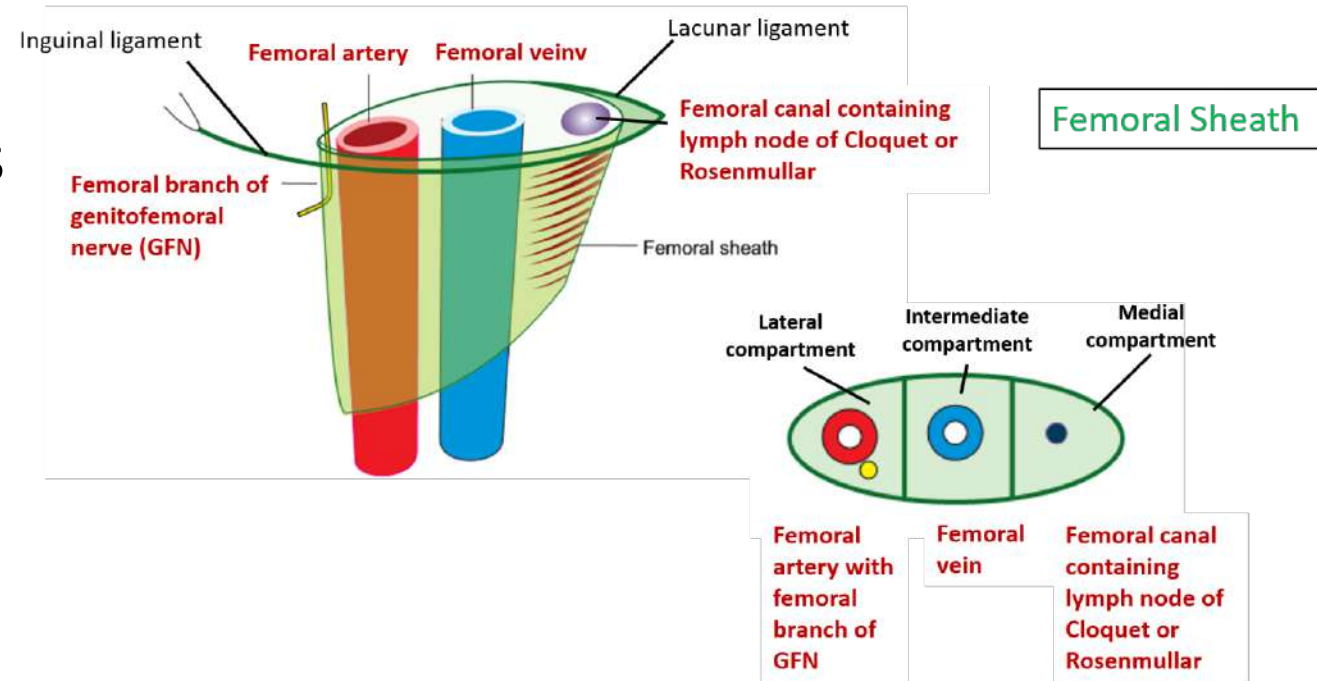


Connection of
femoral canal with
the abdominal
cavity.



Relations of femoral ring:

- **Anteriorly:** inguinal ligament
- **Posteriorly:** superior ramus of pubis
- **Medially:** lacunar ligament
- **Laterally:** Femoral vein

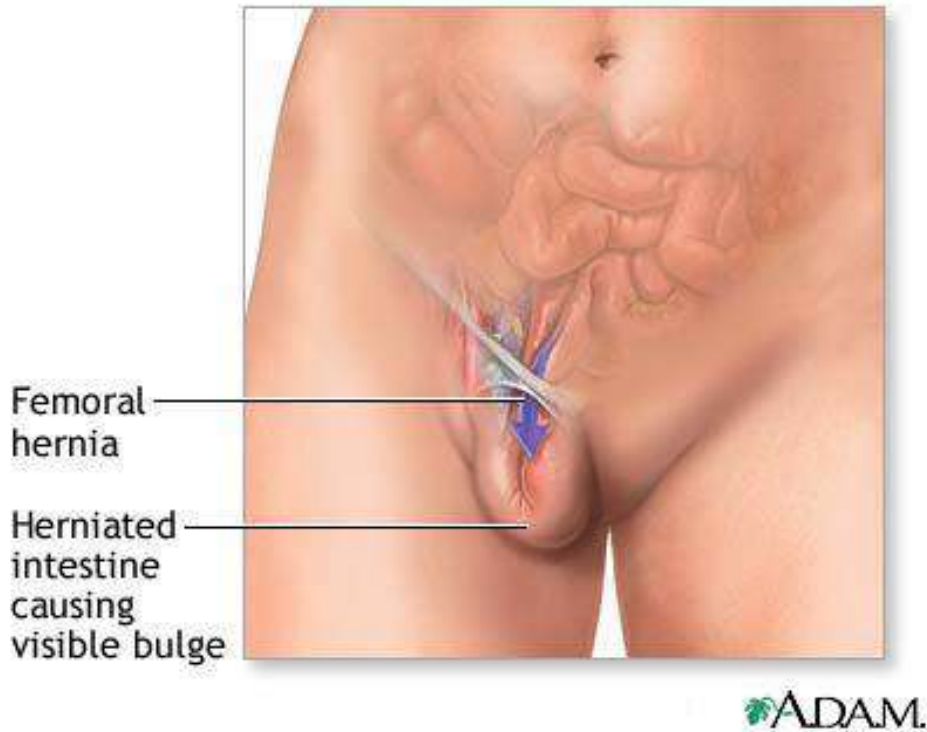


Applied anatomy

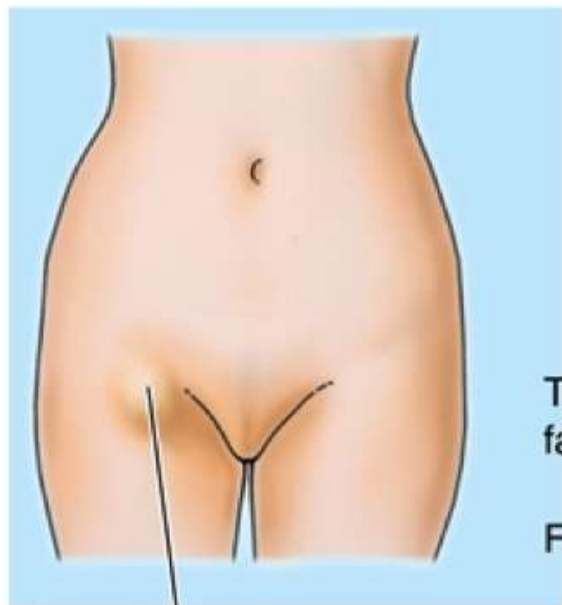


- The femoral canal is surgically important as its abdominal opening (femoral ring) forms **a weak part in the abdominal wall.**
- Femoral sheath is adherent to the walls of femoral vessels, however femoral sheath is not adherent to the wall of small lymph vessels,-----so femoral ring, is a potentially weak point in the lower abdomen. **It is the site for femoral hernias.**

Femoral hernia



- A protrusion of peritoneum, could be forced down the femoral canal, carrying with it a piece of small intestine.
- More common in **females**.
- A mass is **inferolateral to pubic tubercle**.



(A) Early stage

Transversalis fascia

Fascia lata

Femoral hernia

Inguinal ligament

Femoral artery and vein

Saphenous opening

Anterior view

Iliopsoas muscle and fascia

Femoral nerve

External iliac artery and vein (cut)

Inferior epigastric artery

Intestine

Aberrant obturator artery

Femoral ring

Lacunar ligament

Pubic symphysis

Pubic tubercle

Femoral sheath

Femoral hernia

Great saphenous vein



Adductor Canal (Sub- sartorial canal)Hunter

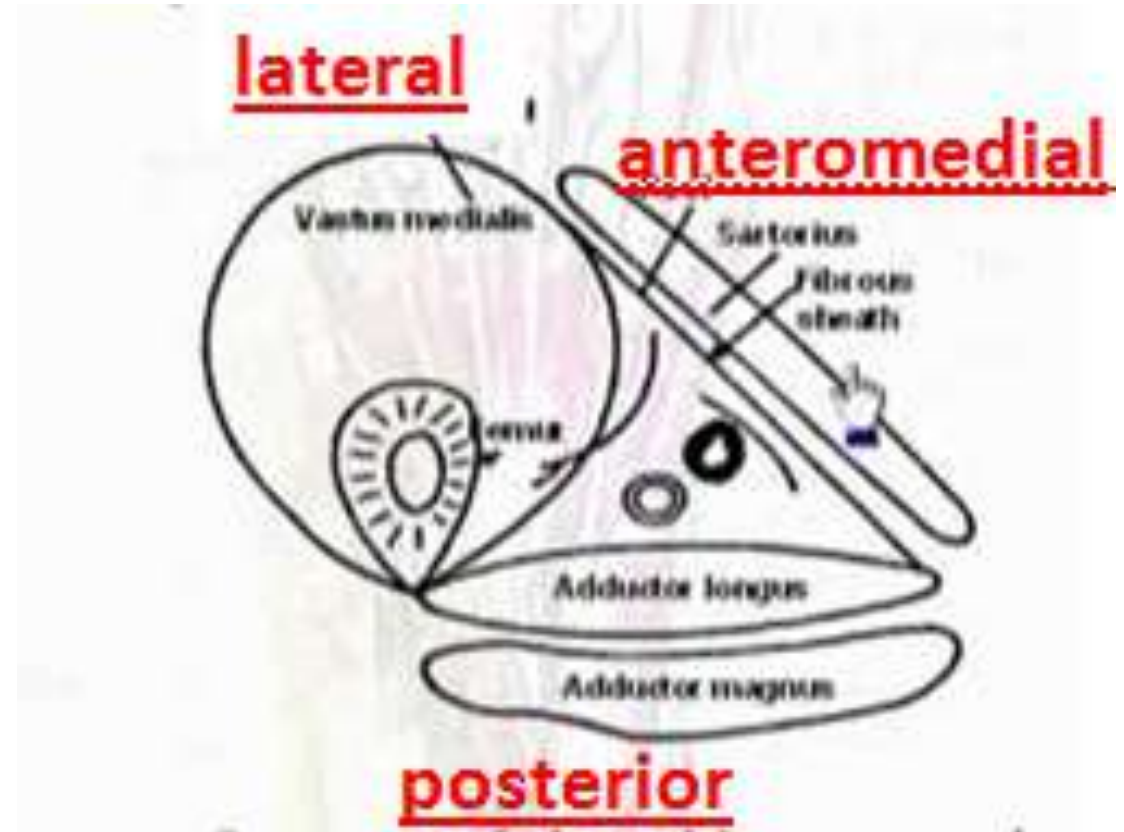
المحطة الثانية

- ❑ **Site:** It is intermuscular fascial cleft situated on the medial aspect of the Middle third of the thigh under sartorius muscle.
- ❑ **Beginning (above):** at the apex of femoral triangle.
- ❑ **End (below):** at adductor hiatus.
- ❑ **Importance????** Delivering femoral vessels to **popliteal fossa** , through **adductor hiatus**, where they become popliteal vessels.



❑ Boundries(walls):

- In cross section adductor canal is **triangular** & has 3 walls;
- Anteromedial wall (Roof): formed by **sartorius muscle & fascia**.
- Posterior wall(floor): formed by **adductor longus & magnus**.
- Lateral wall: formed by **vastus medialis** (because the canal is medial to vastus medialis)



Left adductor canal



rectus femoris

vastus intermedius

vastus medialis

vastus lateralis.

Site of
adductor
canal

Left Lower Limb

□ Content:

1-Terminal part of femoral artery

2-Femoral vein

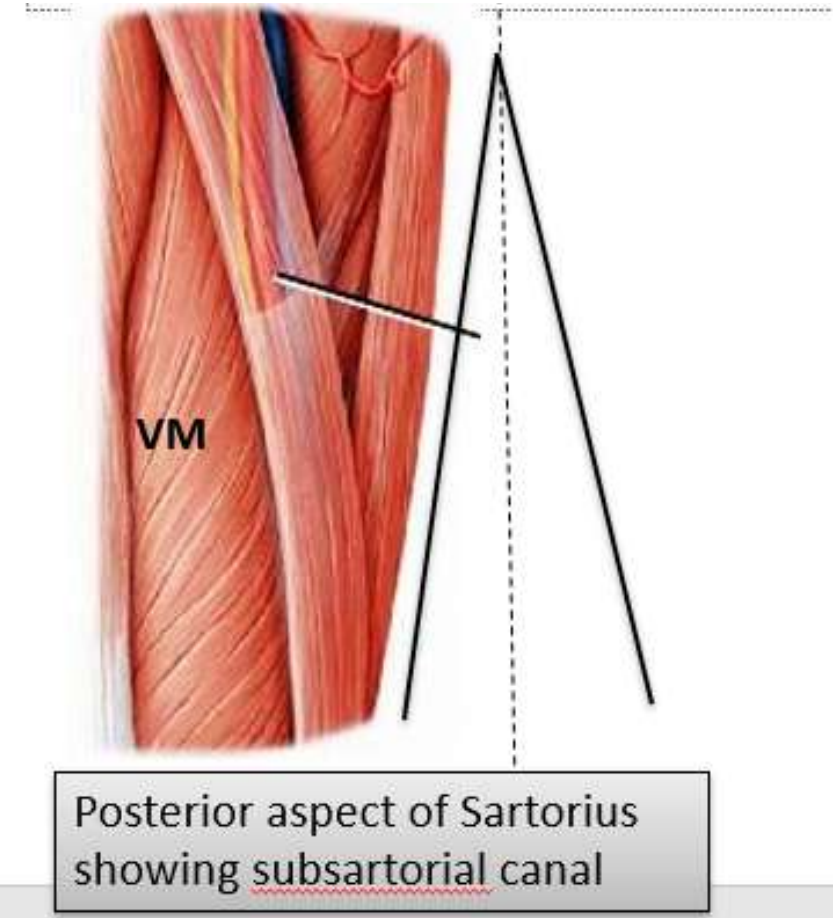
3-Deep lymph vessels

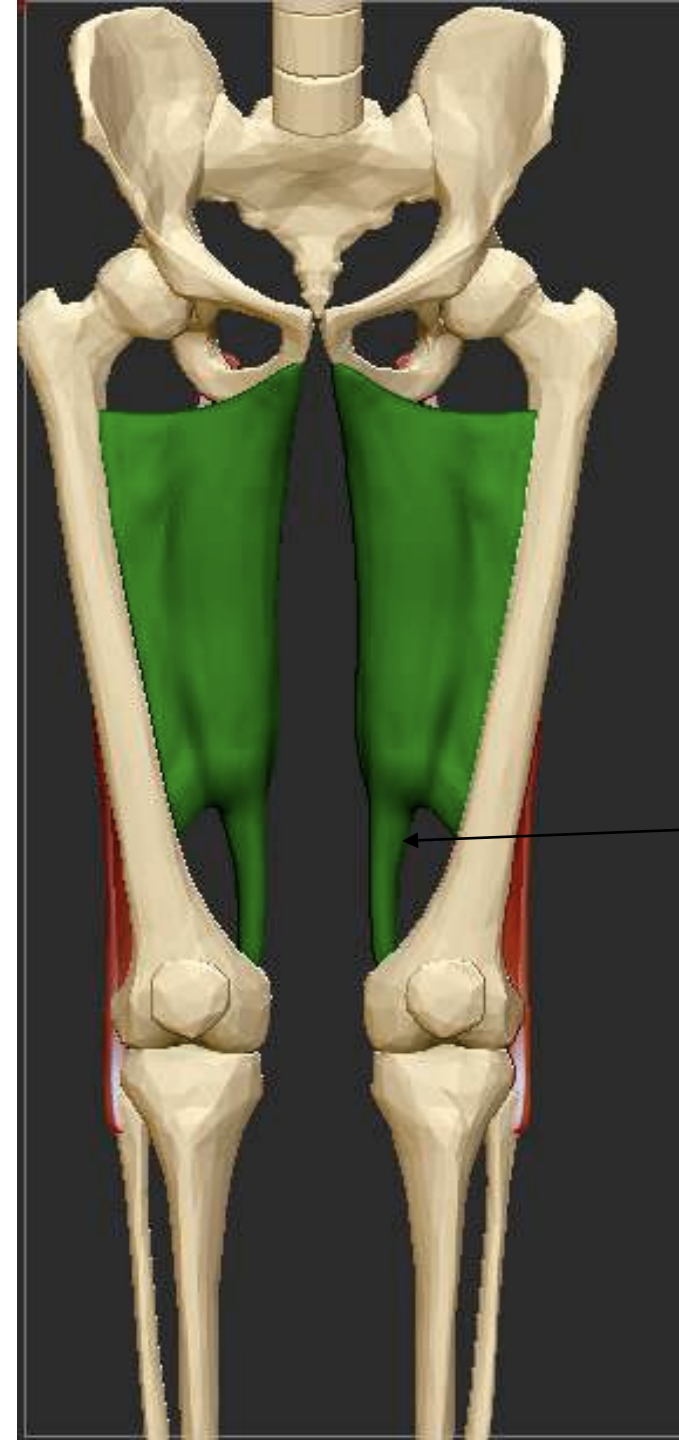
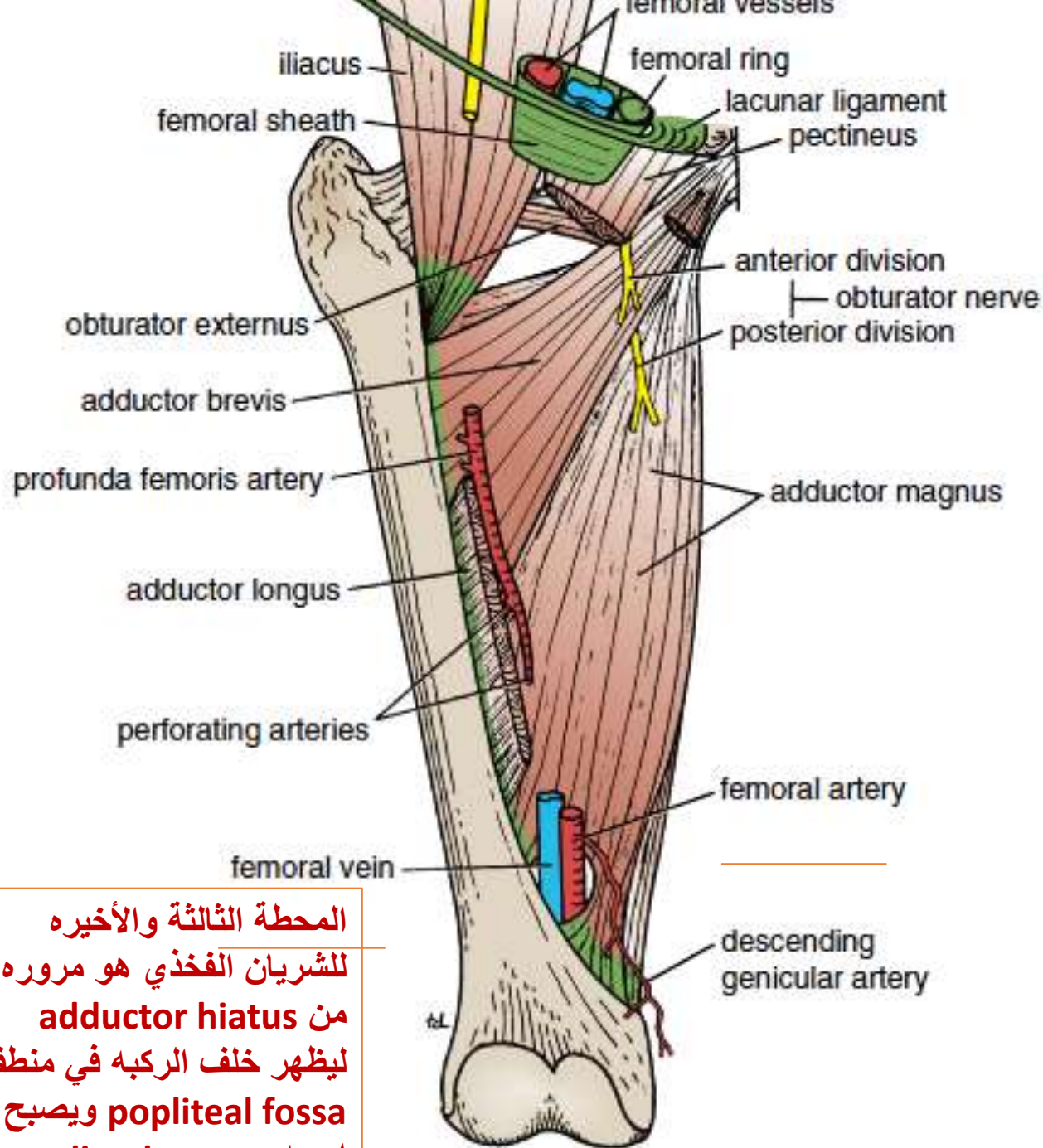
4-Saphenous nerve

5-Nerve to Vastus Medialis

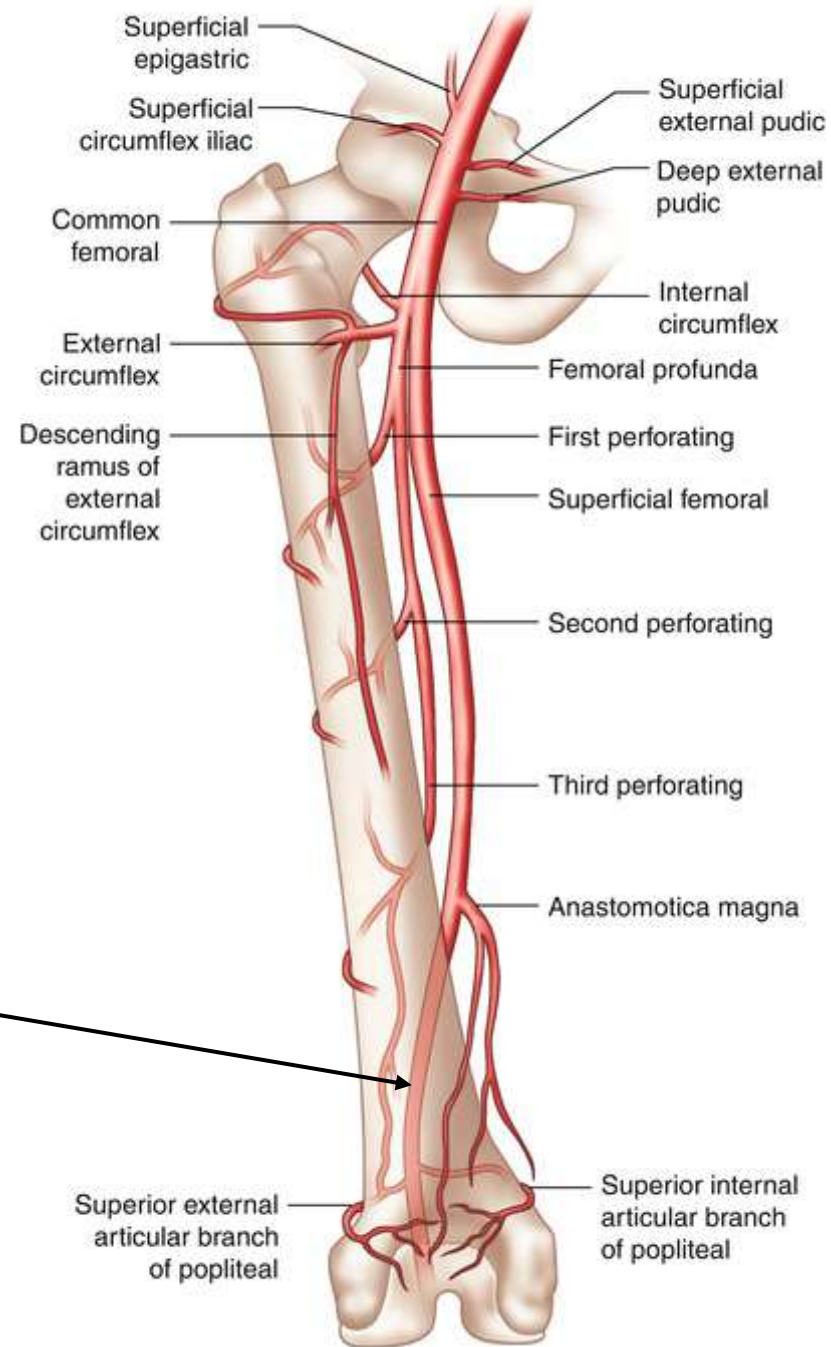
6-Terminal part of Obturator nerve.

Branches of
femoral nerve





المحطة الثالثة والأخيرة
للشريان الفخذي هو مروره
من **adductor hiatus**
ليظهر خلف الركبة في منطقة
popliteal fossa ويصبح
اسمه **popliteal artery**.

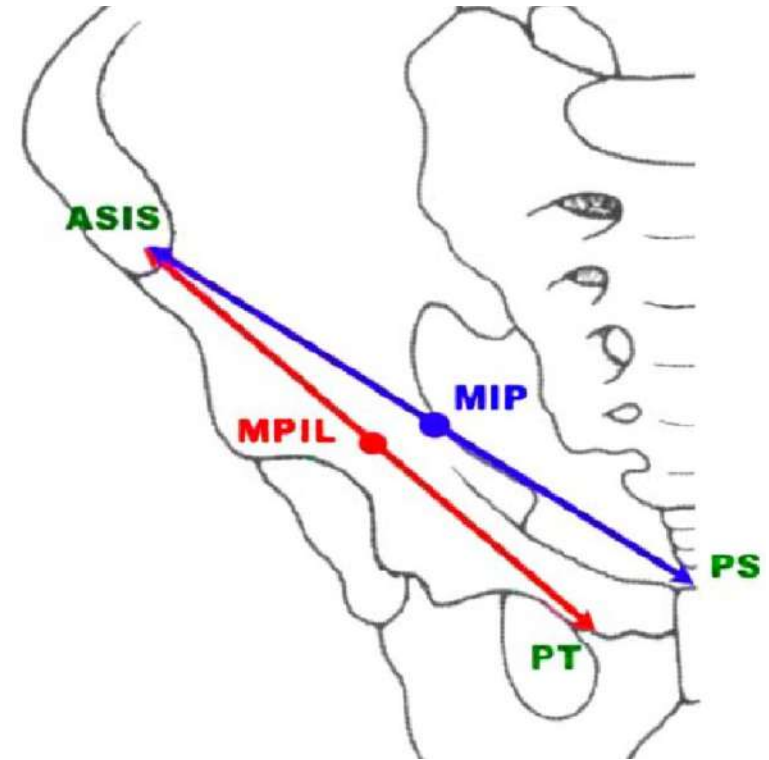


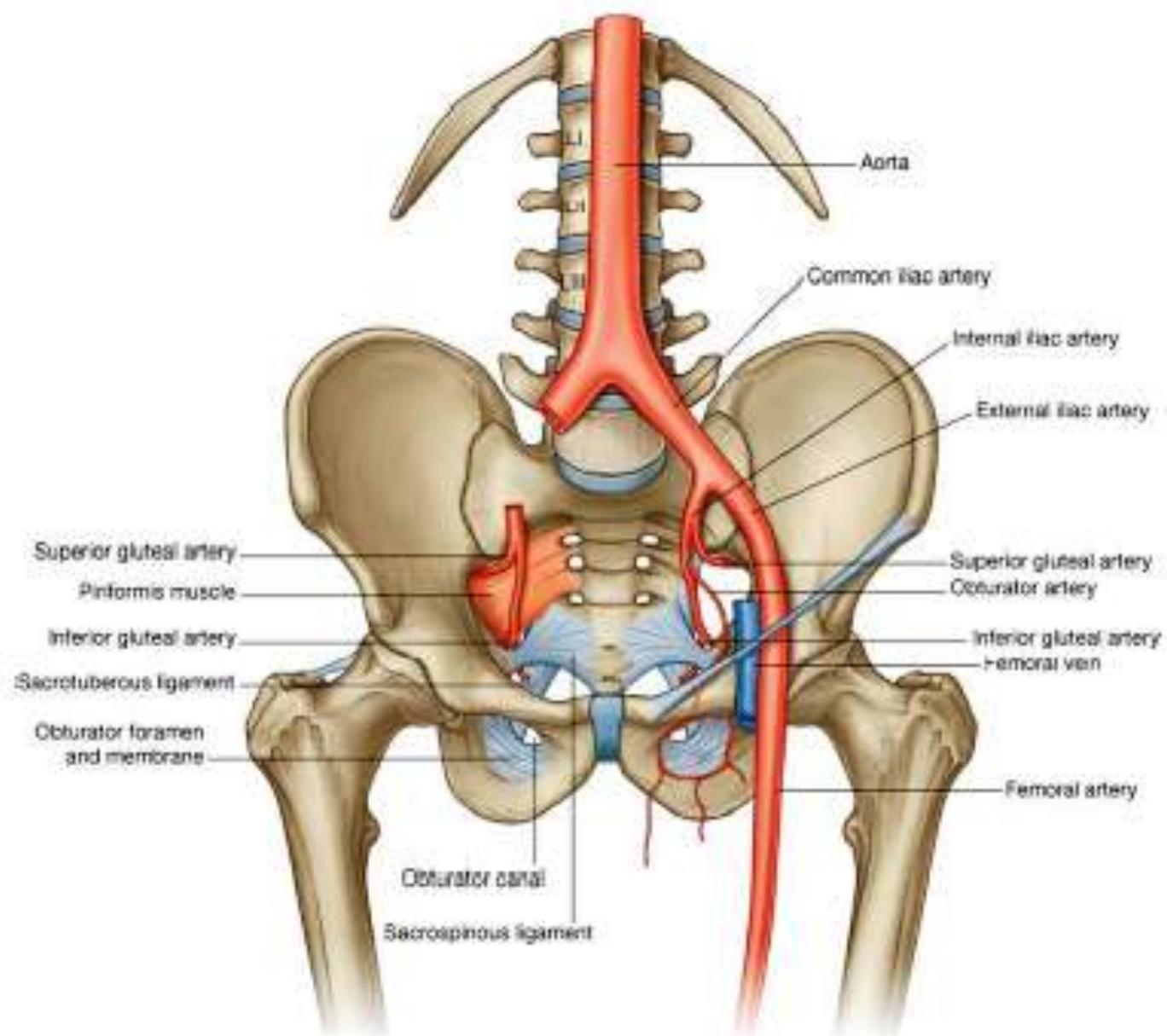
يظهر هنا الشريان
الفخذي بالاحمر
المضلل حيث انه
الان خلف الركبه في
منطقة popliteal
fossa

Blood and nerve supply of anterior
compartment of the thigh

الشريان الفخذي Femoral artery

- ❑ It is the primary artery of the lower limb.
- ❑ **Beginning:** It enters the thigh by passing behind the inguinal ligament (**Midinguinal point (MIP):** midway between the anterior superior iliac spine and the symphysis pubis.) as a continuation of **external iliac artery**.
- ❑ **Course:** 1) In the femoral triangle. 2) in the adductor canal 3) passing through adductor hiatus toward popliteal fossa.
- ❑ **End:** It ends at **adductor hiatus**(in adductor magnus) by entering popliteal fossa as the **popliteal artery**.





☐ Relations of femoral artery:

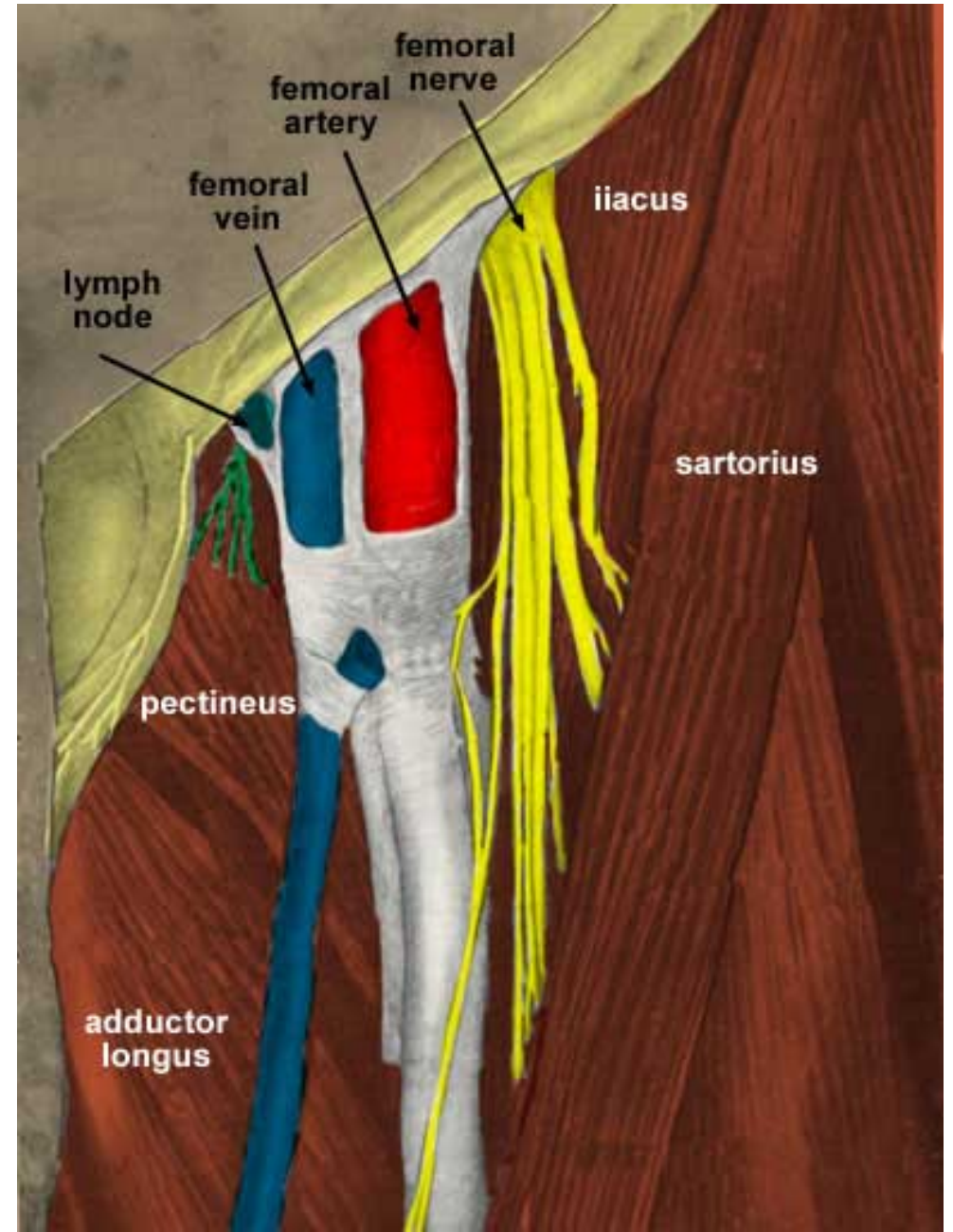
1-Ant: upper part: it is superficial & covered by skin and fascia(**pulsation is palpable**).

Lower part: covered by sartorius, in adductor canal.

2-post: it lies on psoas, pectineus, adductor longus.

3-Medially: Femoral vein.

4-Laterally: Femoral nerve & its branches.



□ Branches 3/3:

s { 1- **Superficial** circumflex iliac artery.—to ASIS

2- **Superficial** epigastric artery—to umbilicus region

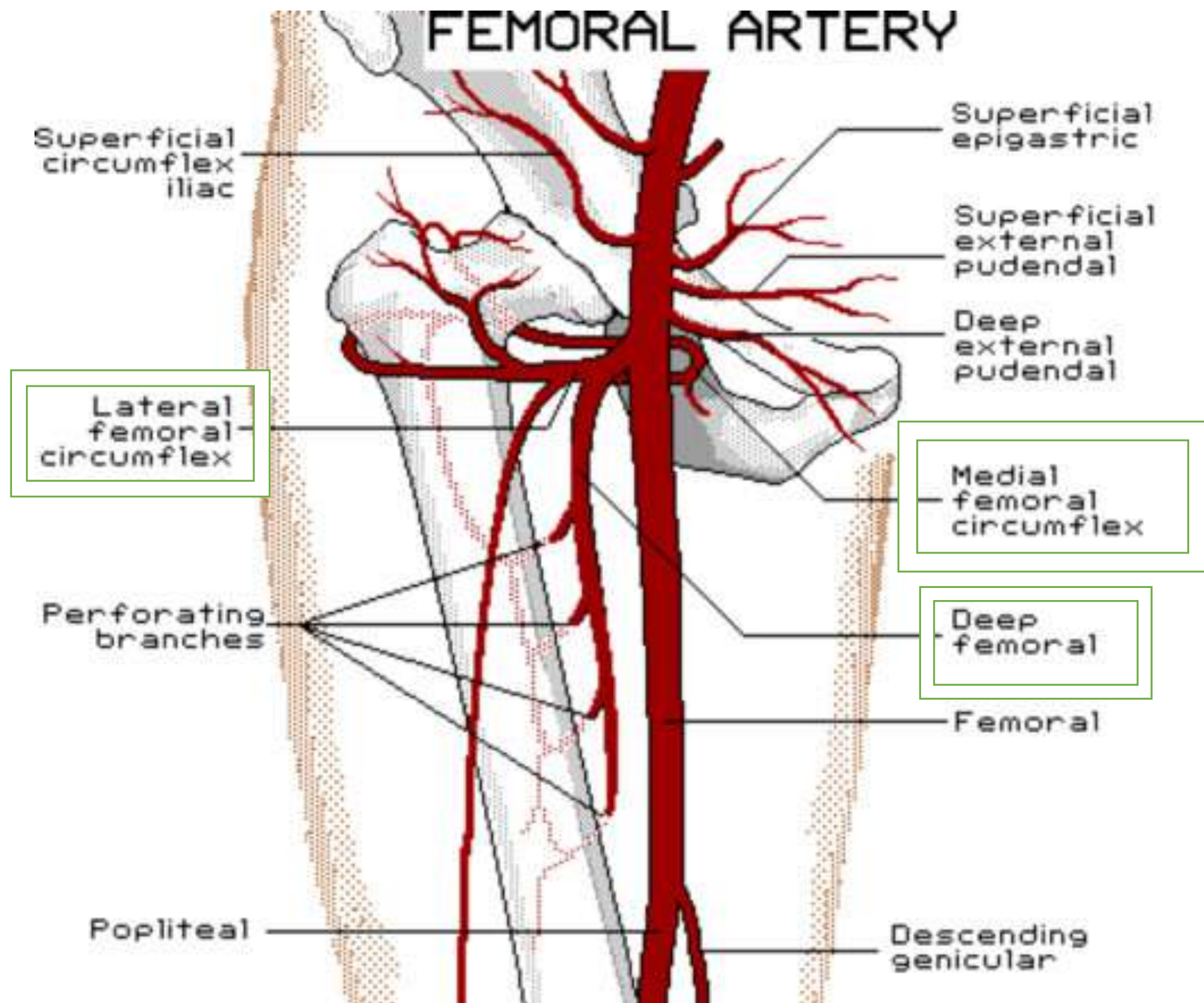
3- **Superficial** external pudendal artery—to skin of scrotum

D { 4- Deep external pudendal artery---to skin of scrotum

5- **Profunda femoris artery (Deep Femoral artery)**

6- Descending genicular artery ;arises near its termination , assists in supplying the **knee joint**.

FEMORAL ARTERY



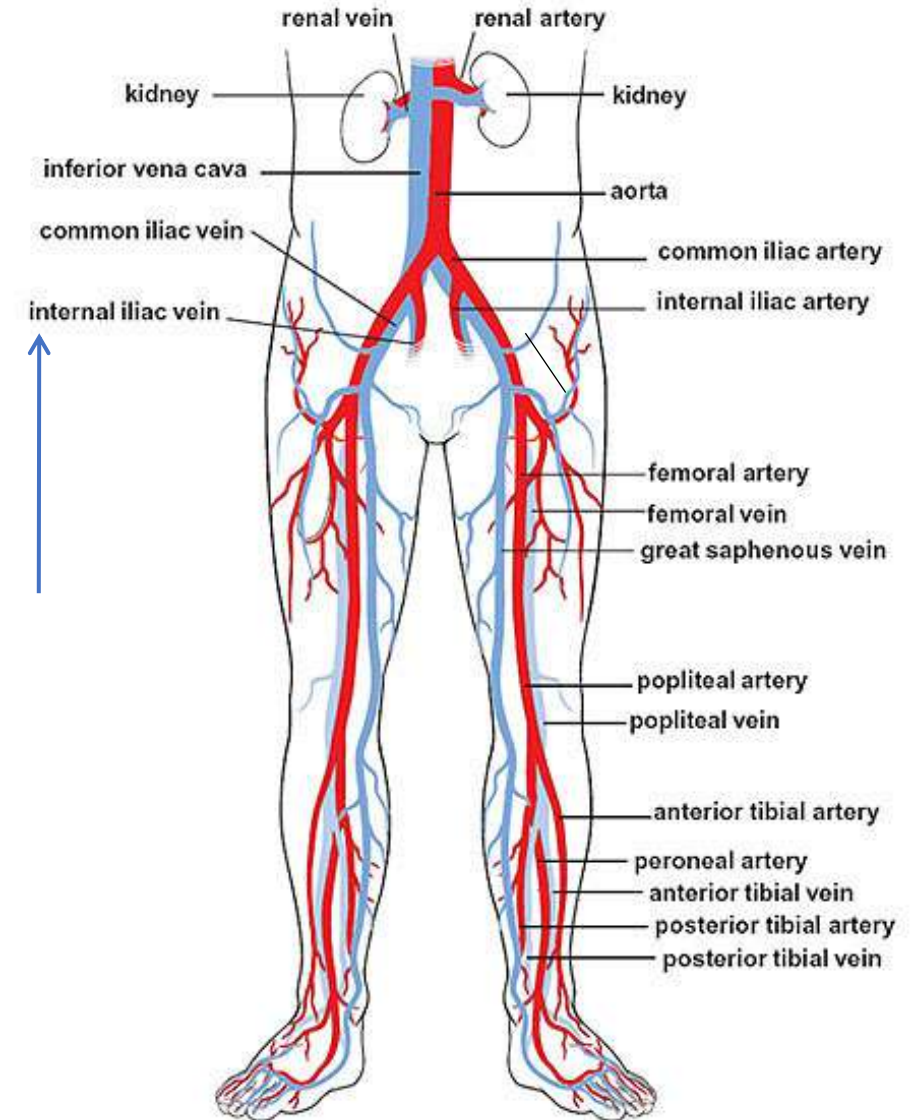
Profunda femoris (deep femoral)artery

- Is the **largest branch** of the femoral artery and the chief artery of the thigh.
- **Origin:** arises from the lateral side of femoral artery in the femoral triangle, 4cm below inguinal ligament.
- **Course:** Passes medially behind femoral vessels& enters the medial fascial compartment of the thigh.
- **End:** by becoming the 4th perforating artery.
- **Branches:**
 - 1- **Medial & lateral femoral circumflex;** at its origin, medial femoral circumflex artery passes backward, supply most **blood to head and neck of femur**. Lateral femoral circumflex; passes laterally, mainly supplies muscles on the lateral side of the thigh. They take part in the formation of cruciate anastomosis.
 - 2- **Four perforating arteries;** three of these arises during its course in the **middle third** of the thigh, wrap around the posterior aspect of the femur. Supply muscles of all three fascial compartments. **The 4th perforating artery is the terminal part of profunda artery.**

❖ الوريد الفخذي: Femoral vein

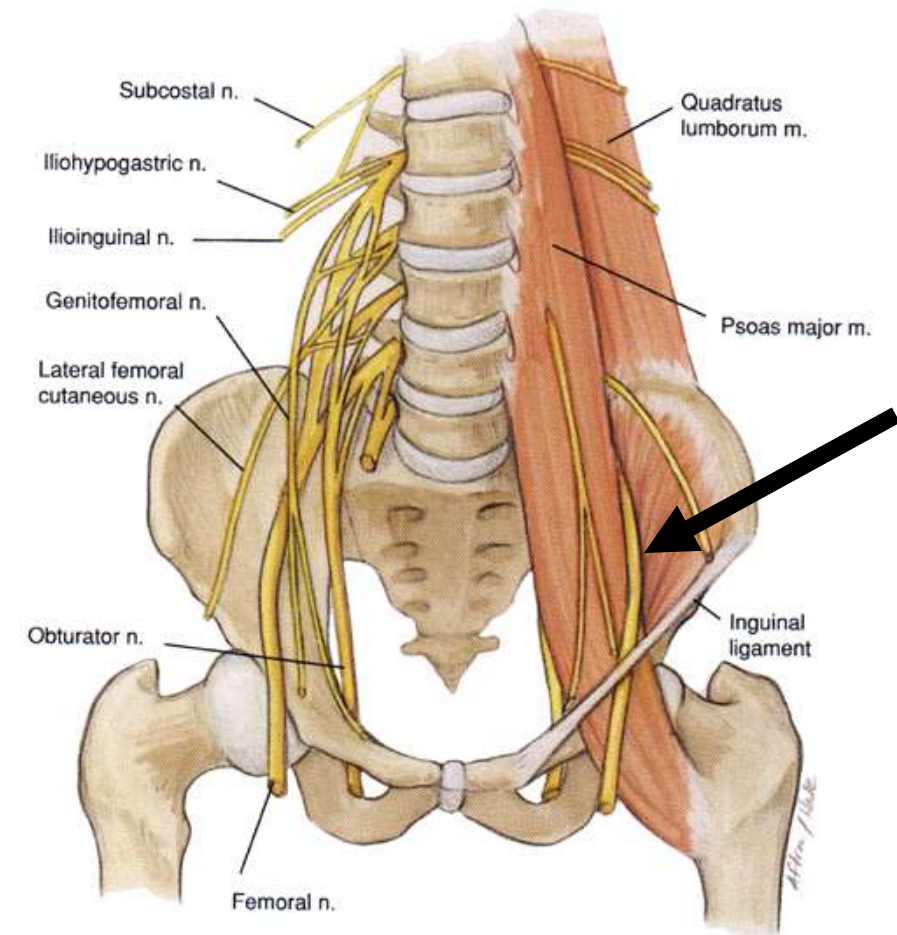
- **Beginning:** Entering the thigh by passing through the opening in the **adductor magnus** as a continuation of the **popliteal vein**.
- **Course:** It ascends through the thigh, lying at first on the lateral side of the artery, then posterior to it, and finally on its medial side.
- **End:** It leaves the thigh in the intermediate compartment of the femoral sheath & passes behind inguinal ligament to become **external iliac vein**.
- **Tributaries:** in the inferior part of femoral triangle, it receives **profunda femoris & great saphenous veins** + other veins that correspond to the branches of the femoral artery.

NB: superficial circumflex iliac vein, superficial epigastric vein & external pudendal veins drain into the great saphenous vein.



Femoral nerve العصب الفخذي

- ❖ Is the **largest branch of the lumbar plexus.**
- ❖ Origin(Root): (L 2,3 and 4)
- ❖ Course:
 - ❖ in the abdomen: It emerges at the lateral border of psoas major.
 - ❖ In the pelvis: It lies in the interval **between psoas and iliacus muscles.**
 - ❖ It passes behind the inguinal ligament(**midpoint of inguinal ligament**) and enters the thigh.
 - ❖ in the **femoral triangle**: It lies **lateral** to femoral vessels and femoral sheath.

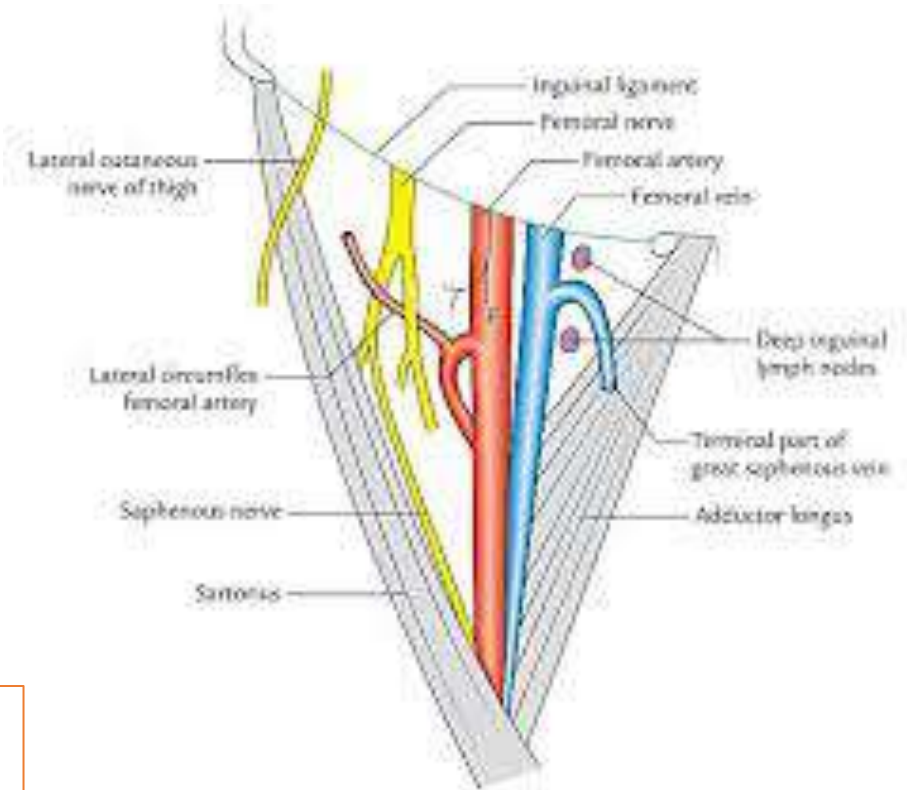


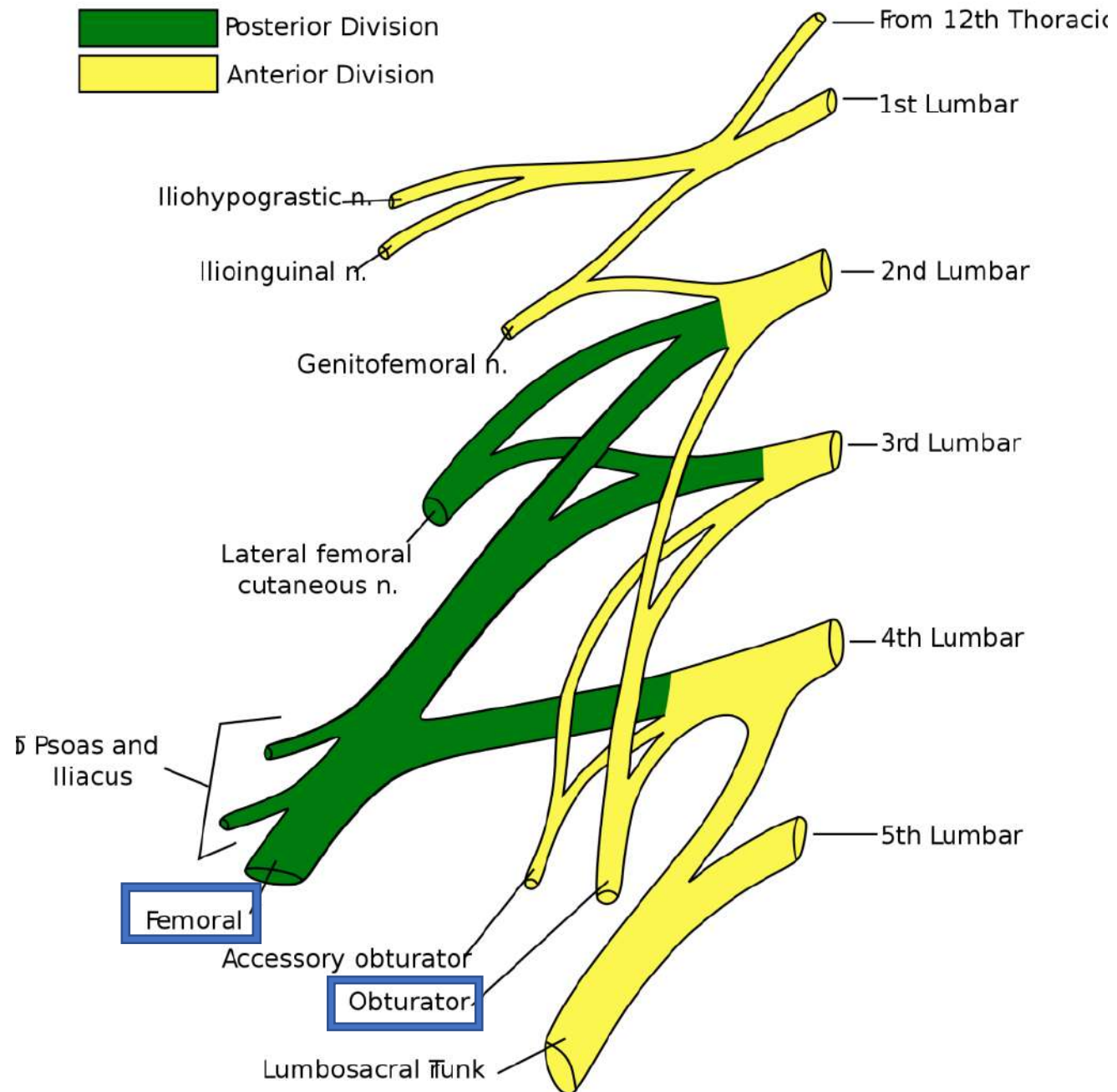
❖ **End:** In the femoral triangle, 4cm (1.5 in) below the inguinal ligament, it terminates by dividing into **anterior and posterior divisions**.

❖ **Supplies:** It supplies all the muscles of the anterior compartment of the thigh **Except Psoas major**.

❖ **Nerve to iliacus:** arises from the femoral nerve (nerve trunk), in the abdomen and supplies iliacus in the abdomen.

Note: Femoral nerve *does not* enter the thigh within the femoral sheath.





Lumbar Plexus

❖ Branches:

a- From the trunk: Nerve to **iliacus**

b-Anterior division: gives off: **2 cutaneous / 2 muscular**

1-Medial cutaneous nerves of the thigh(to skin of medial thigh)

2-Intermediate cutaneous nerves of the thigh (to skin of anterior thigh)

3-Muscular branches to Sartorius.

4-Muscular branches to Pectineus.

c-posterior division: gives off: **1 cutaneous / 1 muscular**

1. Cutaneous: Saphenous nerve.

2. Muscular branches to quadriceps femoris muscle.

Femoral Nerve

Dorsal divisions of ventral rami of L2.3.4 spinal nerves

Branches from Trunk

Nerve to Iliacus

Nerve to Pectineus

Branches from Posterior division

Nerve to rectus femoris (motor)

Nerve to Vastus lateralis(motor)

Nerve to Vastus intermedius(motor)

Nerve to Vastus medialis (motor)

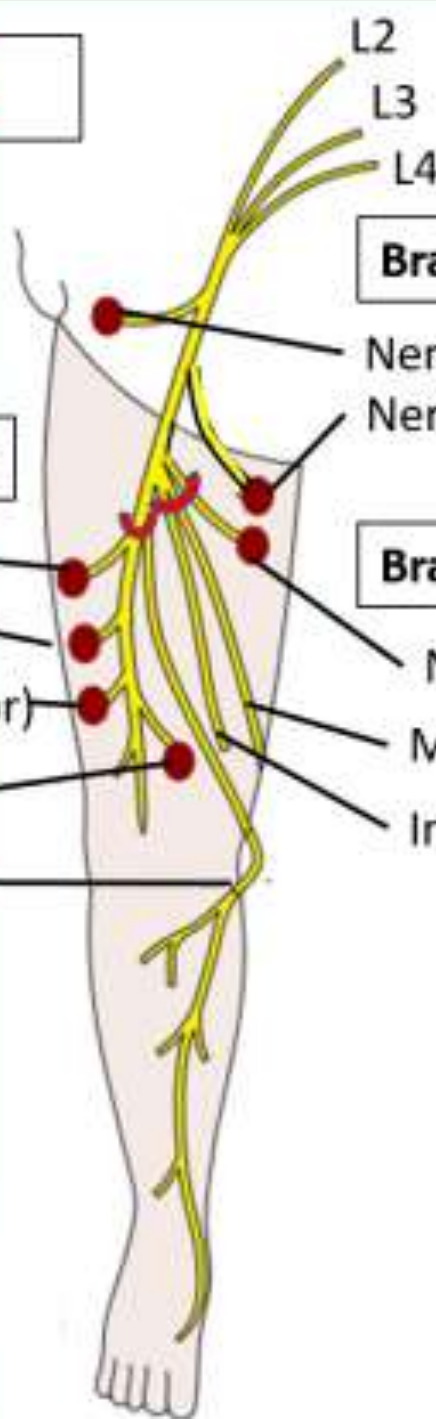
Saphenous nerve (sensory)

Branches from Anterior division

Nerve to Sartorius (motor)

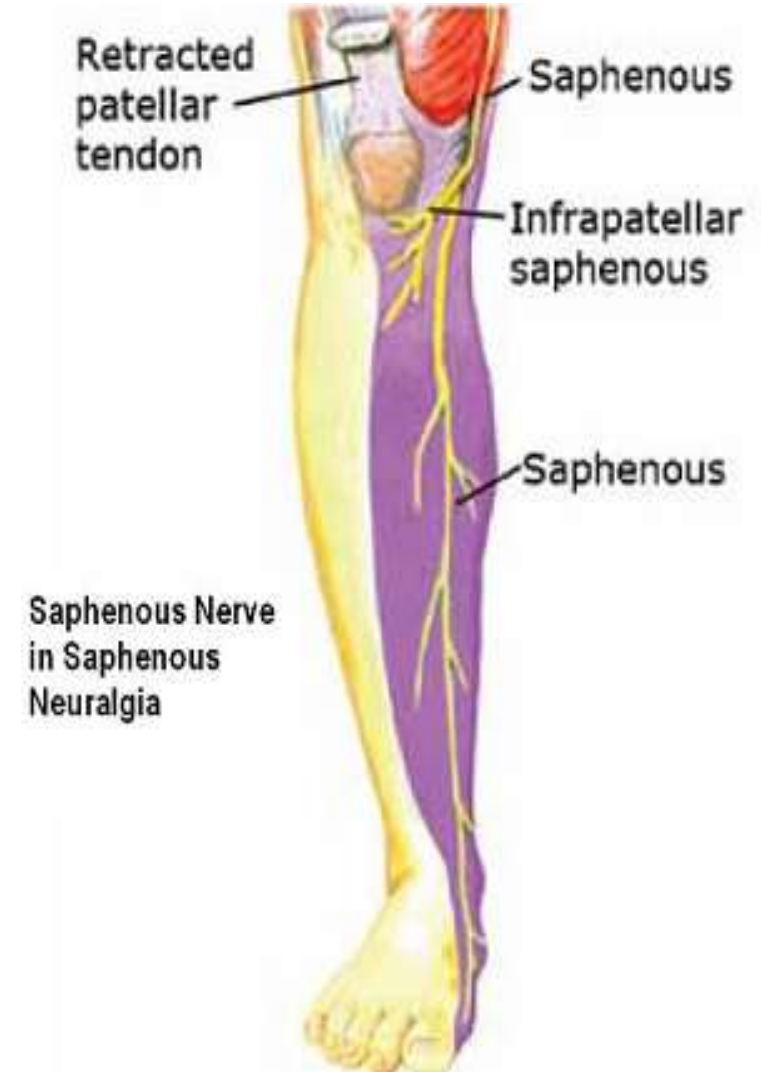
Medial cutaneous nerve (sensory)

Intermediate cutaneous nerve (sensory)



Saphenous nerve

- **longest** cutaneous nerve in the body.
- It accompanies the femoral vessels through adductor canal. Then it become superficial.
- It descends with great saphenous vein on the medial side of the thigh & leg. It passes **in front of the medial malleolus with the great saphenous vein**.
- **It gives an infrapatellar branch** which shares in patellar plexus.
- It supplies **skin on the medial side of the leg and medial part of the dorsum of the foot**.
- **Ends:** along the medial border of the foot in the region of the **ball of big toe(metatarso- phalangeal joint of the big toe)**.



Femoral nerve injury

❑ **Cause of injury:** stab or gunshot wounds

❑ **Clinical picture with complete nerve division(rare):**

motor: quadriceps femoris is paralyzed & the knee cannot be extended.

Sensory: skin sensation is lost over **anterior & medial sides of the thigh, medial leg and the medial border of the foot as far as the ball of the big toe.**



Femoral nerve
does not enter
the thigh within
the **femoral**
sheath.



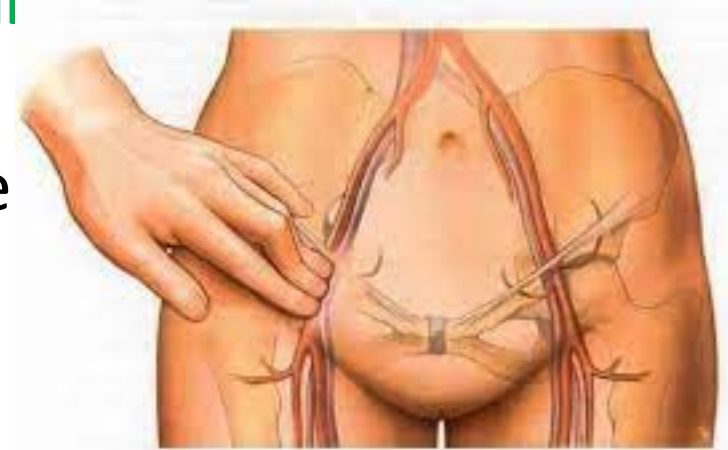
Applied

❑ Palpation, compression & cannulation of femoral artery

- ❑ Person lying in the supine position, femoral pulse may be palpated **1 cm below mid inguinal point (mid-way between the ASIS and the pubis symphysis.)**. If external iliac artery is partially occluded, pulse may be diminished.

❑ Compression of femoral artery;

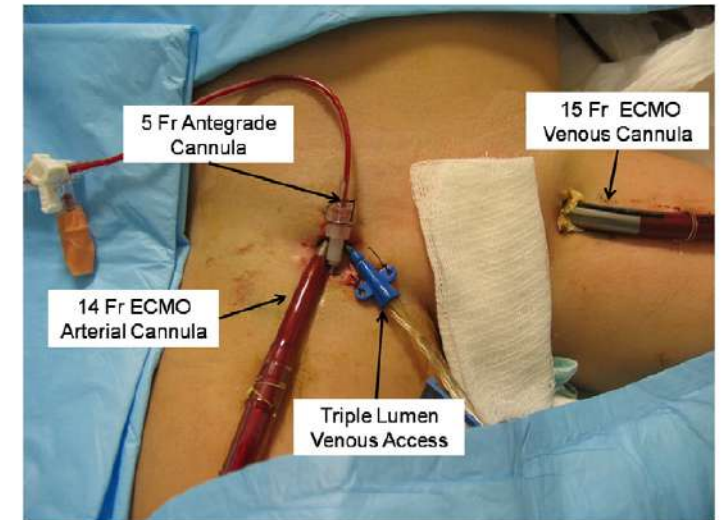
to reduce blood flow to femoral artery & its branches in case of bleeding ,by pressing directly posteriorly **against superior pubic ramus, psoas major & femoral head.**

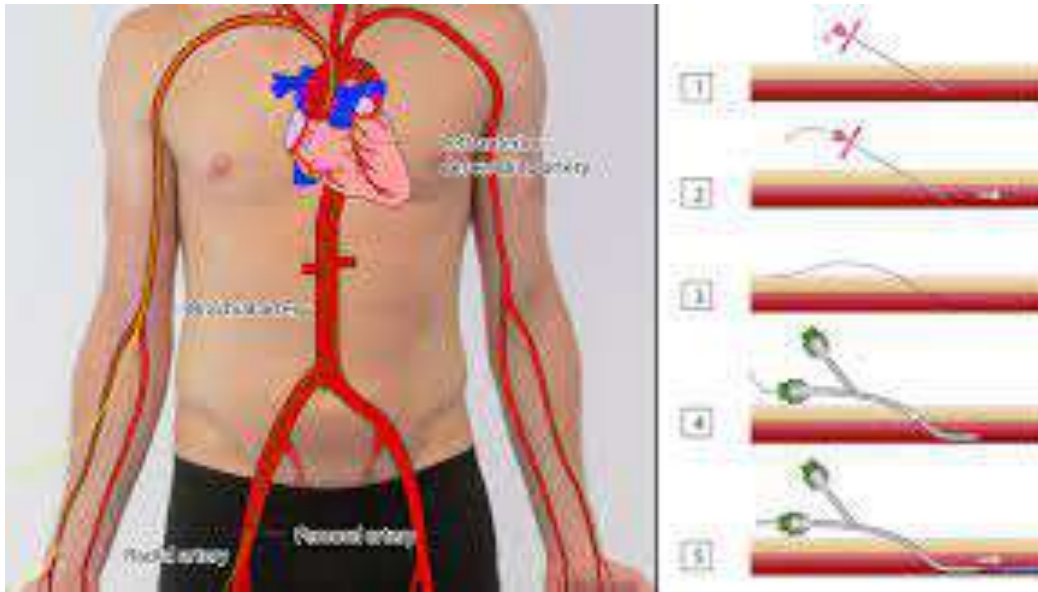


- ***Cannulation of femoral artery:***

- Just inferior to midinguinal point, (**angiography**) femoral artery can be cannulated using a catheter , which passed up to the **external iliac artery, common iliac artery and aorta to left ventricle** of the heart.

- Arterial blood can be taken for blood gas analysis.





Sandbag
applied to
procedure site



After cardiac catheterization

Coronary
cathetrization from
femoral artery

Blood and nerve supply of Medial compartment of the thigh

Blood supply of the medial compartment of the thigh

1-Profunda femoris artery :

mentioned before

Profunda femoris vein :

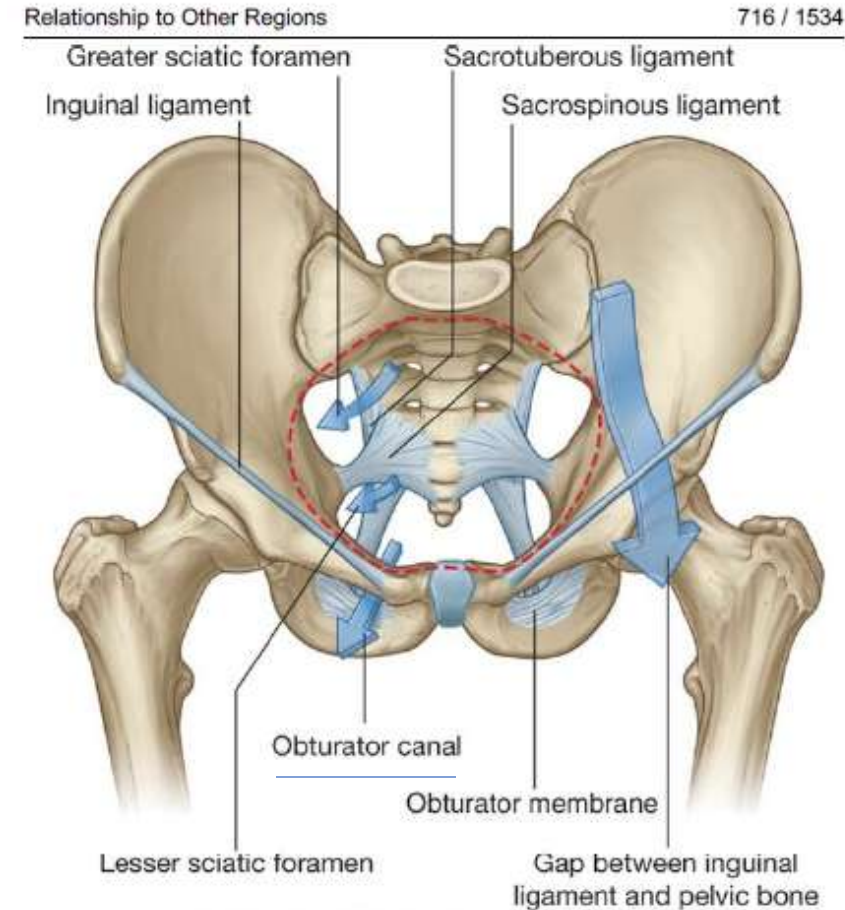
Receives tributaries correspond to branches of artery. Drains into femoral vein.

2-Obturator artery

- Is a branch of the **internal iliac artery**.
- Passes through obturator canal (i.e in the upper part of obturator foramen) with the obturator nerve.
- In the medial compartment of thigh it divides into **anterior & posterior branches**. It gives muscular & articular branches to hip joint.

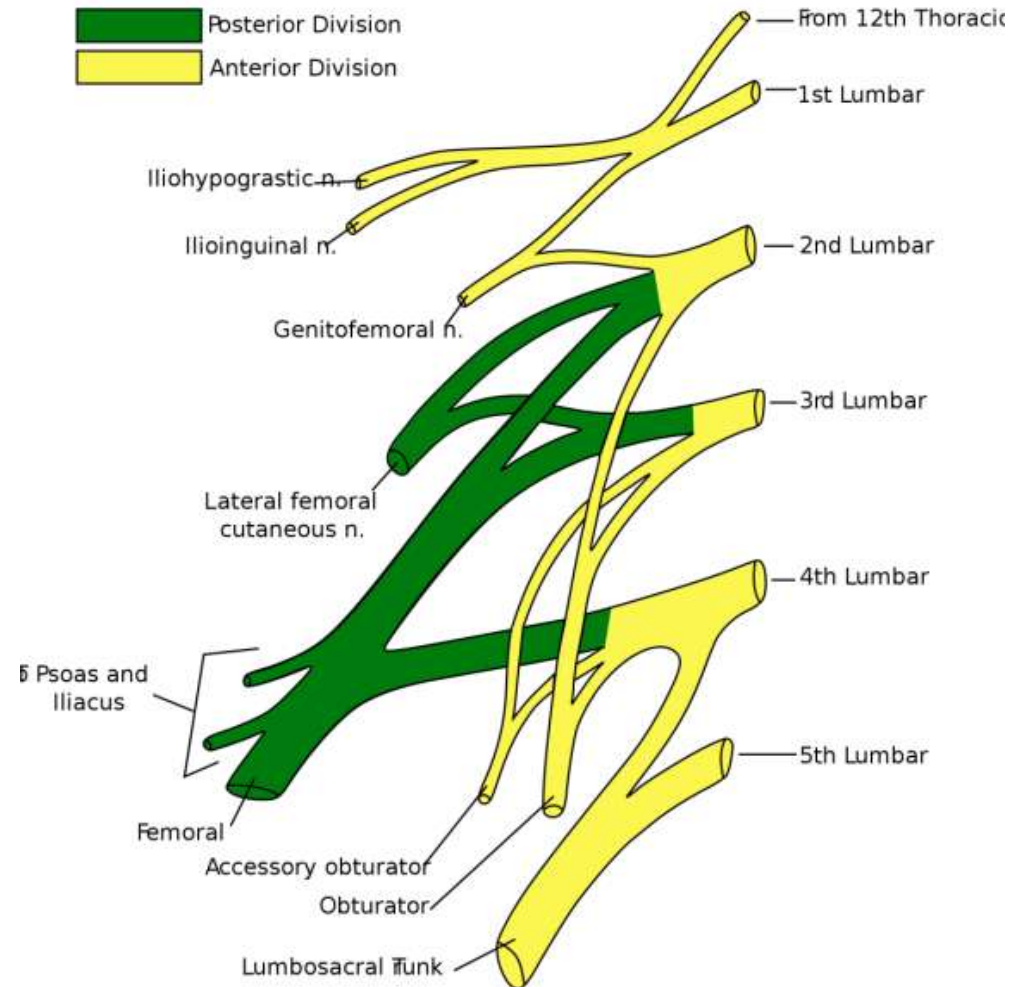
Obturator vein:

Receives tributaries correspond to branches of artery & it drains into **internal iliac vein**.



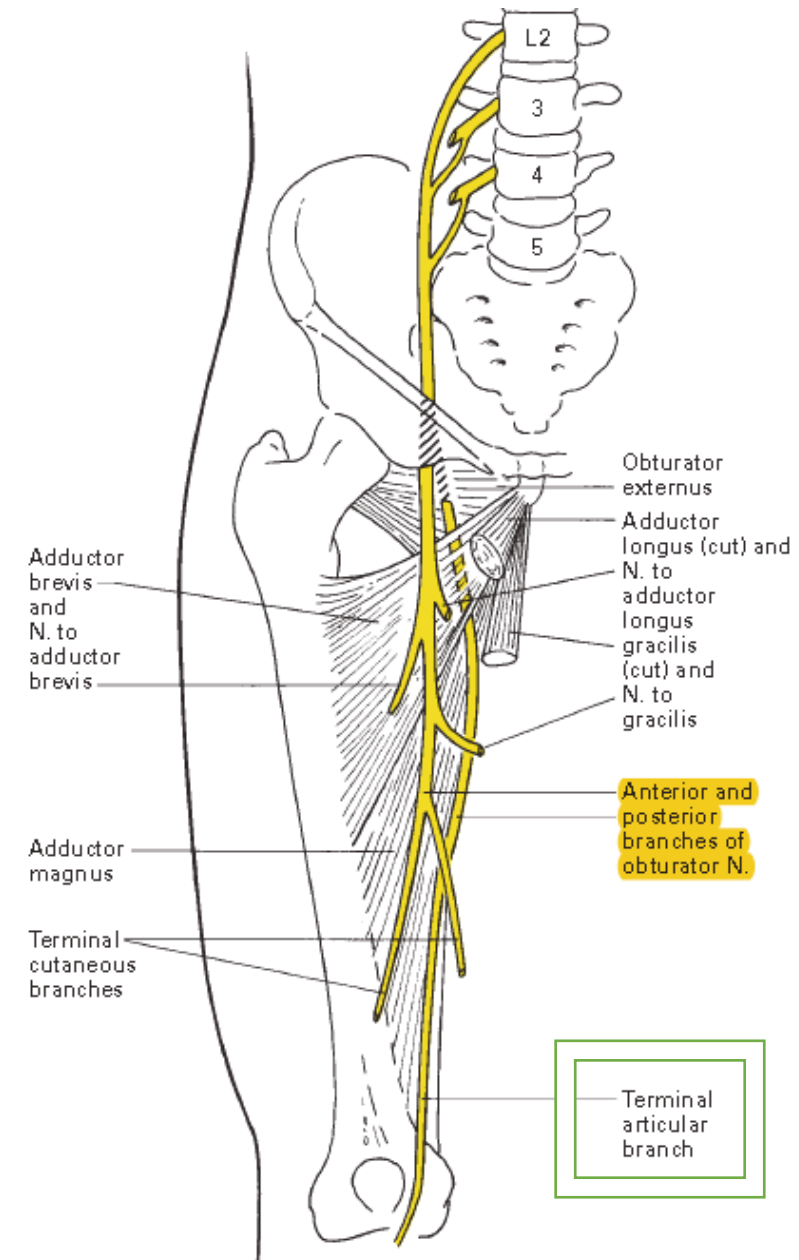
Nerve supply of the medial compartment of the thigh.

Lumbar plexus



□ Obturator nerve

- **Origin:** arises from **lumbar plexus**(L2 ,3 and 4).
- **Course:**
 - ❖ It emerges on the **medial border** of the **psoas muscle** within the abdomen.
 - ❖ It reaches the thigh by passing through the **obturator canal**.
- **End:** in the upper part of obturator foramen divides into **anterior and posterior divisions**. It leaves adductor canal with the femoral artery by passing through the adductor hiatus. The nerve terminates by supplying the knee joint(post.division).



A) Branches of anterior division:

a-Muscular branches to : Gracilis, adductor brevis & adductor longus & occasionally pectineus.

b-Articular branches to : hip joint.

c-Cutaneous branches to : skin of the middle third of the medial side of the thigh.

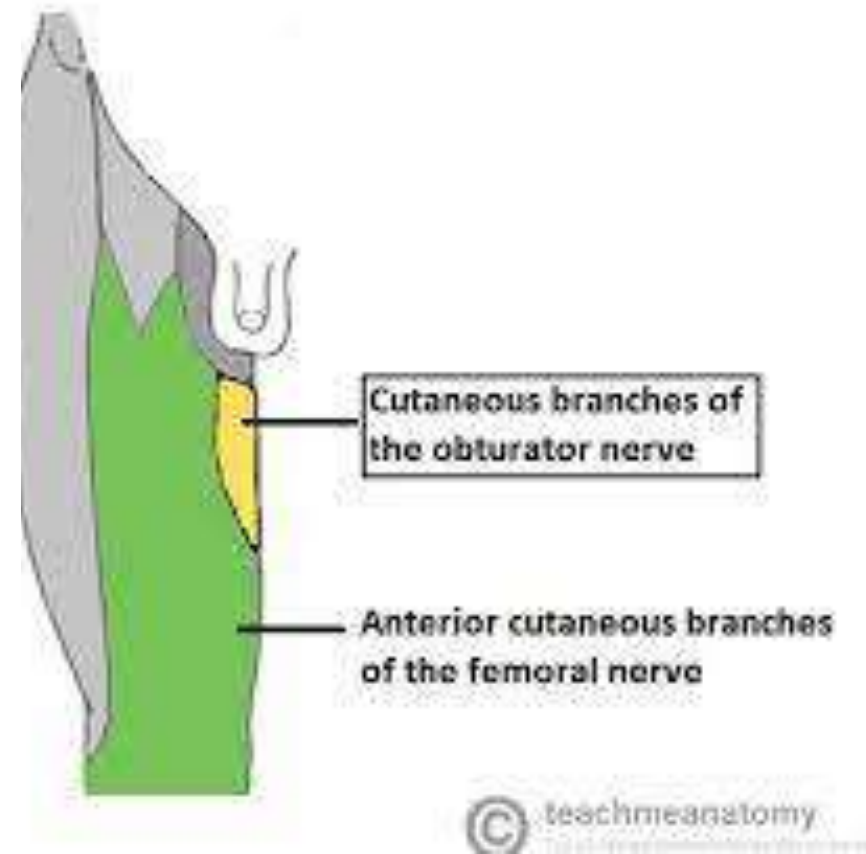
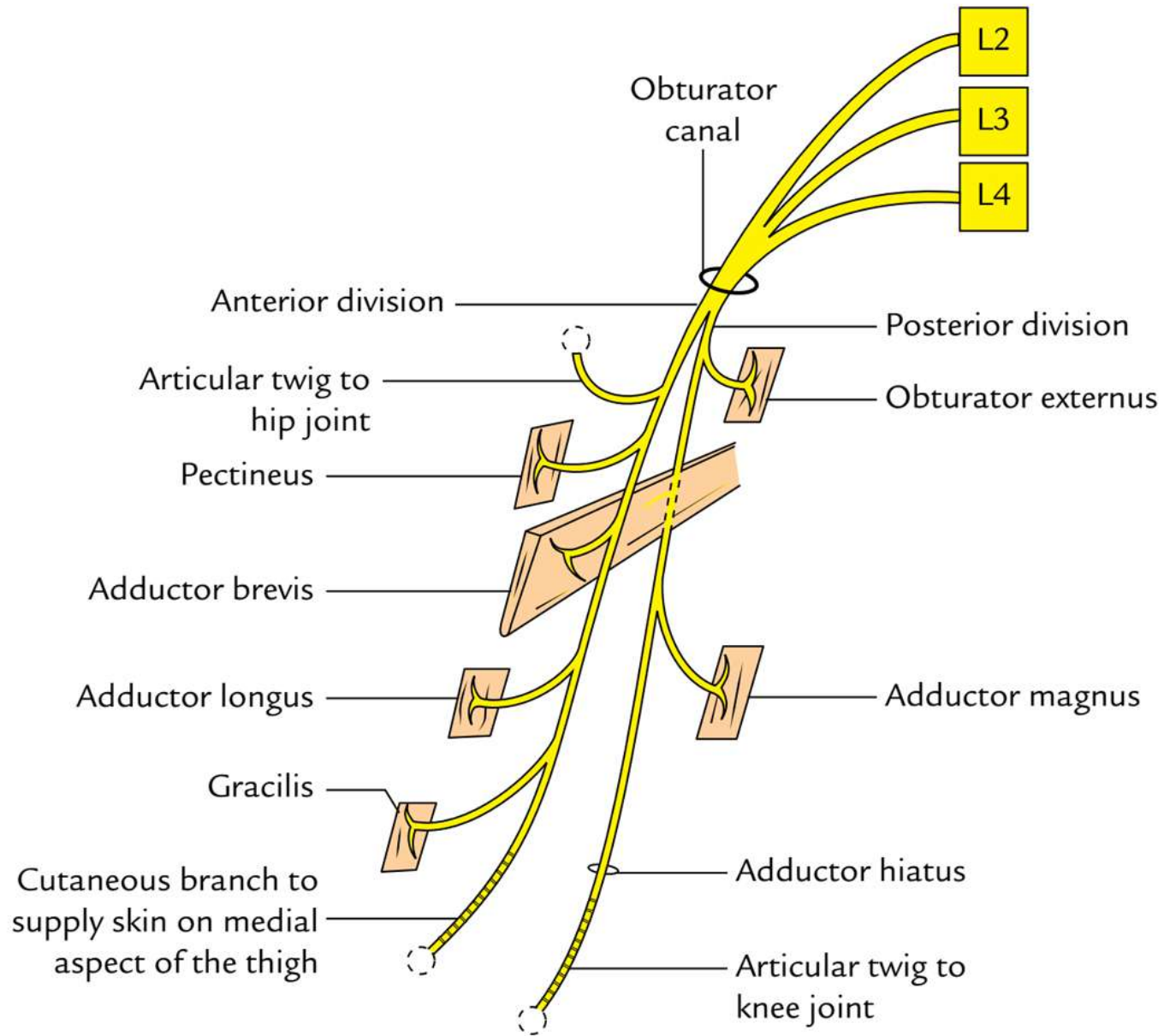
NB: It terminates as a small nerve supplies the femoral artery.

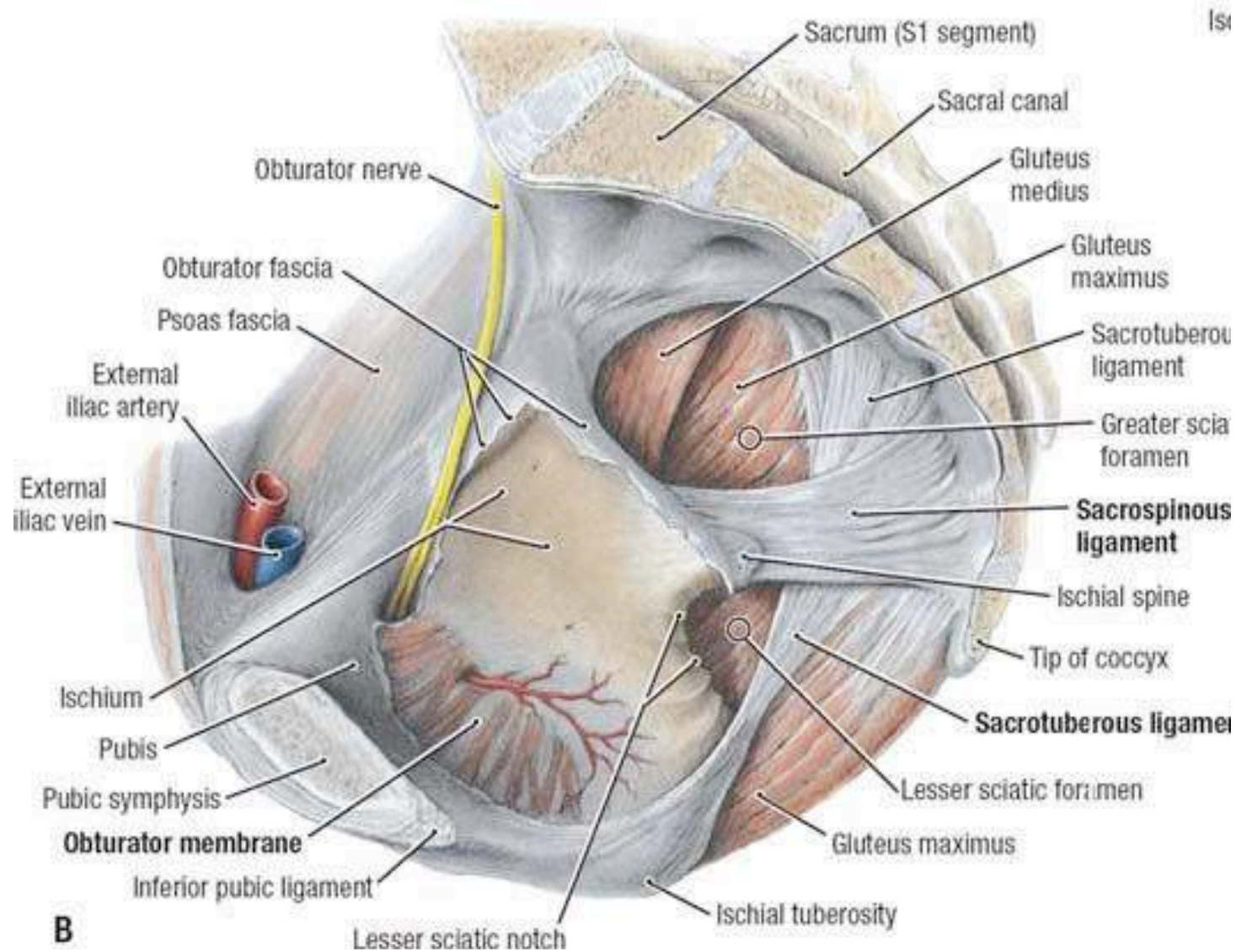
B) Branches of posterior division:

a-Muscular branches to: obturator externus, adductor part of adductor magnus and occasionally adductor brevis.

b- Articular branches to: Knee joint

NB: It terminates by descending through adductor hiatus to supply the knee joint.





Obturator nerve in the obturator foramen

- ❑ All the muscles of the medial compartment of the thigh are supplied by obturator nerve *Except* hamstring portion of A.magnus by sciatic nerve.
- ❑ All the muscles of the medial side of the thigh, act on hip joint only Except gracilis, which also act on knee joint causing flexion of the leg at knee joint (i.e act on both hip& knee joint).

❑ All of the muscles of the medial compartment of the thigh are inserted into posterior surface of femur bone except

Gracilis (tibia, SGS) , **obturator externus** (greater trochanter) and **hamstring portion of adductor magnus** (adductor tubercle of femur).



- ❑ All of the adductors are inserted into **femur** bone **Except** Gracilis inserted in **tibia** (upper/medial surface).
 - ❑ Action of all adductors is **adduction of the thigh at hip joint** & assist in lateral rotation.
 - ❑ hamstring portion of A.magnus *extends thigh*
- At a hip joint.*
- ❑ Obturator externus *causing lateral rotation of the thigh at the hip joint.*

- ❑ In general; Three muscles attached to linea aspera of femur; from above downward; ***pectinus, adductor brevis, adductor longus.***
- ❑ ***in general; Muscles originate from pubis:***
 - 1-Sup. pubic ramus: Pectinus.***
 - 2-Body of pubis: Adductor longus.***
 - 3-Inf. Pubic ramus: Gracilis, adductor brevis and adductor magnus.***
- ❑ Muscles originate from ramus of ischium are:
Gracilis and adductor magnus.

- ❖ Rectus femoris and adductor magnus have two heads of origin.
- ❖ In general three muscles act on hip and knee joint: 1- Sartorius 2- Rectus femoris 3-Gracilis.



(Formative assessment)

1-All the following muscles are supplied by femoral nerve **except?**

- A. Iliacus
- B. Psoas major
- C. Pectineus
- D. Sartorius

2-The subsartorial (adductor) canal is located in?

- A. Upper third of thigh
- B. Middle third of thigh
- C. Lower third of thigh
- D. Leg

3-Which of the quadricep femoris muscles performs extension as well as flexion?

- A. Vastus lateralis
- B. Vastus medialis
- C. Vastus intermedius
- D. Rectus femoris

4-Medial boundary of femoral triangle is formed by?

- A. Adductor magnus
- B. Adductor longus
- C. Sartorius
- D. Iliopsoas

5-The lateral wall of adductor canal is formed by?

- A. Sartorius
- B. Vastus lateralis
- C. Vastus medialis
- D. Adductor longus



النجاح ليس صدفة. إنه عمل
شاق، ومثابرة، وتعلم، ودراسة،
وتصحية، والأهم من ذلك كله،
حب ما تفعله.